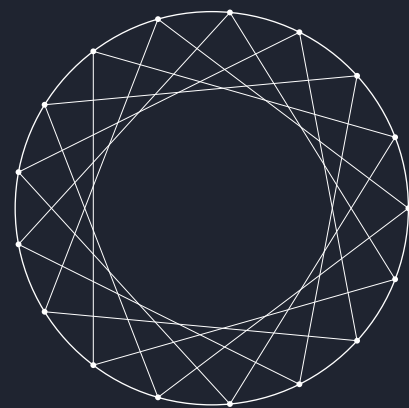


OLYMPIAD PROBLEM SOLVING

A dense mini-textbook & cheat sheet for the IMO and the Putnam



HOW TO USE THIS

read once, then keep beside you

This is not a list of formulas. Each section is organised around **what the problem looks like** (the *signal*) and **what to reach for** (the *tool*). The olympiad is won less by knowing theorems than by recognising, within five minutes, which three of them the problem is quietly asking for.

- **Boxes** state results precisely: the hypotheses matter; half of olympiad errors are misapplied hypotheses.
- **Left-rule notes** are heuristics, tricks and the “why” behind a tool.
- **Red notes** are traps that cost points.
- **Worked examples** (framed, with a *Lesson*) show the tool in action on real competition problems; try each before reading the solution.
- ► **SIGNAL:** lines tell you when a tool is likely to be the intended one.

■ Colour key

- Strategy & meta-heuristics
- Algebra: inequalities, polynomials, FEs
- Number theory
- Combinatorics & graph theory
- Euclidean geometry
- Putnam: analysis, linear algebra, algebra, probability

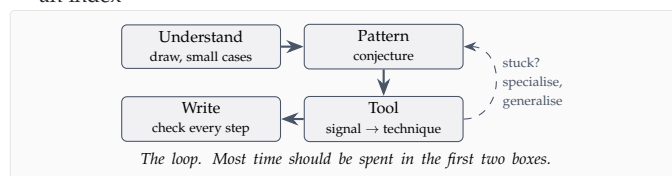
■ IMO vs Putnam at a glance

	IMO	Putnam
Format	2×4.5h, 3 problems each	2×3h, 6 problems each (A, B)
Scoring	0–7 per problem	0,1,2,8,9,10
Allowed maths	“pre-calculus” only	anything: calculus, linear algebra, groups
Difficulty curve	P1/P4 easy, P3/P6 brutal	A1/B1 easy, A6/B6 brutal
Grading	partial credit for real progress	nearly all or nothing

Putnam scoring: a complete solution to 1 problem (10) beats serious progress on 4 (often 0–2 each). Finish and *write up* easy problems perfectly before hunting hard ones.

■ Contents

1. Strategy: the problem-solving loop, a heuristic catalogue, proof shapes, training method
2. **Algebra:** classical inequalities & how to choose; SOS, pqr/uvw , tangent-line; polynomials; functional equations; sequences
3. **Number theory:** divisibility, orders, LTE, Zsigmondy, quadratic residues, Diophantine techniques, p -adic valuations
4. **Combinatorics:** pigeonhole, invariants, extremal principle, double counting, bijections, generating functions, graphs, games, probabilistic method
5. **Geometry:** configurations, angle chasing, power of a point, transformations, inversion, coordinates (complex, barycentric, trig)
6. **Putnam toolkit:** limits & series, integrals, inequalities in analysis, linear algebra, abstract algebra, probability
7. Further worked examples across all areas; anatomy of a write-up
8. Reference tables: identities, sums, small numbers
9. Formula compendium: identities, inequalities, trigonometry, number theory, counting, geometry, calculus, linear algebra, constants
10. Practice & plan: a problem ladder by technique, a 12-week schedule, an index



■ The five-minute triage

1. Compute small cases ($n = 1, \dots, 6$) *by hand*. Guess the answer.
2. Identify the extremal/equality case. It tells you which inequality, which construction, which configuration.
3. Ask: what is *invariant*? what is *forced*? what is *maximal*?
4. Name the domain: is this secretly a graph? a polynomial? a lattice? a probability?
5. Reverse-engineer: what would make the problem trivial? Can you reduce to that?

I STRATEGY

heuristics that transfer across every domain

■ Pólya's questions, sharpened

- **Have I seen the *unknown* before?** Same target (e.g. "prove n is a square") suggests the same finishing move.
- **Drop a condition.** What survives? Which condition does the real work? If you never used a hypothesis, you are probably wrong.
- **Specialise.** Degenerate triangle, n prime, all variables equal, $x = 0$, f linear.
- **Generalise.** Strengthening the claim can make induction work (*inventor's paradox*).
- **Work backwards.** From the conclusion to something you can prove.
- **Auxiliary object.** A point, a polynomial, a graph, a potential function, a generating function.

■ Universal techniques

Induction variants

Strong (assume all $k < n$). **Forward-backward** (Cauchy): prove for 2^k , then $n \Rightarrow n-1$. **Structural** on trees/graphs: remove a leaf/vertex/edge. **Strengthen the hypothesis** when the step fails. **Infinite descent**: a minimal counterexample produces a smaller one.

Extremal principle. Consider the largest / smallest / leftmost / shortest object. Its extremality yields an inequality no other object satisfies. ► **SIGNAL**: "prove there exists", finite configurations, "some point sees all others".

Invariants & monovariants. Find a quantity preserved (mod m , parity, colouring sum, determinant) or strictly monotone and bounded (so the process terminates). ► **SIGNAL**: any "operation repeated" problem.

Contradiction via counting. If a structure didn't exist, count something two ways and exceed a bound.

Smoothing / mixing variables. Replace (x, y) by $(\frac{x+y}{2}, \frac{x+y}{2})$ (or by $(x+y, 0)$) and show the objective moves the right way; finish when all equal (or at the boundary).

Symmetry breaking. WLOG $a \geq b \geq c$ (only if the statement is symmetric; for cyclic ones, only WLOG $a = \max$).

■ When you are stuck

1. Rewrite the problem in *another language*: algebra $\leftrightarrow$ geometry, sets $\leftrightarrow$ graphs, sequences $\leftrightarrow$ polynomials.
2. Solve a weaker version: fewer variables, specific n , bounds off by a constant.
3. Search for a counterexample in earnest. The failure shows why it's true.
4. List every fact derivable in one step; problems setters rarely include useless data.
5. Try the opposite answer (for "find all" problems, the "obvious" family may be incomplete).

A CATALOGUE OF HEURISTICS

Grouped by the question each one answers. Scan the list when stuck; one of them is usually the door.

■ "What is this problem really about?"

- **Change the representation.** Numbers $\rightarrow$ points on a line, sets $\rightarrow$ binary strings, permutations $\rightarrow$ cycle diagrams, sequences $\rightarrow$ polynomials or generating functions, relations $\rightarrow$ graphs, sums $\rightarrow$ areas, probabilities $\rightarrow$ areas.
- **Name the object.** Give a symbol to the quantity everyone keeps mentioning ($S = a + b + c$, $d = \gcd$, $k = \text{the ratio}$). Naming makes relations visible.
- **Find the hidden structure.** A group acting? A metric? A partial order? A bipartite graph? A linear map?
- **Reformulate the goal.** "Show x is an integer" $\leftrightarrow$ " x is a root of a monic integer polynomial"; "two segments are equal" $\leftrightarrow$ "a point is on a perpendicular bisector".

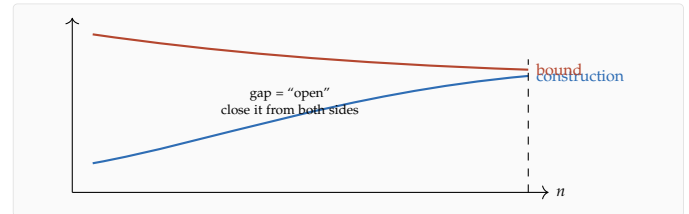
ily may be incomplete).

■ "Find all" problems

Two halves, both mandatory: (i) the claimed objects work (verify!), (ii) nothing else does. Losing the easy half costs a point at the IMO.

■ Answer-guessing for bounds

For "find the max of f " or "the least n such that", you need a **bound** and a **construction attaining it**. Start with the construction; its shape tells you which inequality is tight.



■ Write-up discipline

- State the answer first. Define every symbol. Number claims (*Claim 1*, *Claim 2*) so a partial solution scores.
- Every "clearly" is a potential lost point. Check boundary cases ($n = 1$, equal variables, degenerate figures, 0).
- In geometry, configuration issues (" P lies between A and B ") must be addressed or avoided with directed angles.
- For the Putnam, write complete sentences; graders punish gaps harshly.

■ Common signals $\rightarrow$ first tool

Signal	Try
$a^n \pm b^n$, $p^k \mid \cdot$	LTE, orders, Zsigmondy
$x^2 + y^2$, $x^2 - dy^2$	$\mathbb{Z}[i]$, Pell, Vieta jumping, QR
symmetric n -variable inequality	AM-GM, SOS, uvw , Jensen, tangent line
$abc = 1$	$a = \frac{x}{y}$ substitution, AM-GM, ln
$f(x+y)$, $f(xy)$	plug 0, 1, $x = y$, injectivity, Cauchy
$\sum \binom{n}{k} \omega^k$	roots-of-unity filter
"colour the board"	chessboard/mod- k colouring invariant
"at least two ... coincide"	pigeonhole
set system, $ A_i \cap A_j $	double count, linear algebra over $\mathbb{F}_2$
many cyclic quads	radical axis, Miquel, inversion
tangent circles	homothety, inversion
lim of recursive seq.	monotone+bounded, fixed pt, Stolz-Cesàro
$\int_0^a f(x) dx$ with symmetry	$x \mapsto a - x$
matrix with ± 1 , 0/1	det mod 2, eigenvalues, rank bound
"expected number"	linearity + indicators
game with positions	P/N positions, strategy stealing, pairing

■ "How do I get started?"

- **Data first.** Tabulate $n = 1, \dots, 8$. Look at differences, ratios, prime factorisations, binary expansions. Sequence recognition solves "find $f(n)$ " questions.
- **Draw an accurate diagram** (ruler, compass). Wrong diagrams make false conjectures; accurate ones reveal collinearities, concyclicities, tangencies.
- **Solve the $n = 2$ or $n = 3$ case completely.** Then ask which step used smallness.
- **Try a related, easier problem:** fewer constraints, a continuous version, the "infinite" version.

■ "What must be true?"

- **Necessary conditions.** Before constructing, list conditions any answer satisfies (parity, size, modular constraints). Often they pin the answer.
- **Look at the extremes:** largest element, first time something happens, boundary of a region, leaves of a tree.
- **Local-to-global:** prove something about every small window (two consecutive terms, adjacent cells), then glue.
- **Counting forces existence:** if total $>$ capacity, something doubles up.

■ “How do I go from here to there?”

- **Intermediate target.** Find a statement strictly between hypothesis and conclusion. Most olympiad solutions are 2–4 lemmas.
- **Work backwards** from what the final line must look like.
- **Strengthen the claim** so it can be induced on (Ex. prove a bound for all $k \leq n$, or an identity with an extra parameter).
- **Isolate the obstruction:** find what breaks naive approaches; the fix is often the key lemma.

■ “What can I exploit?”

- **Symmetry** (reduce cases; or break it by an ordering).
- **Parity / modularity** (odd-even, colourings, weights mod m).
- **Convexity** (maxima at endpoints; minima at critical points).
- **Monotonicity** (once something happens it keeps happening).
- **Discrete continuity:** an integer quantity that changes by at most 1 per step takes every intermediate value.
- **Linearity:** expectations, determinants in each row, sums of indicators.

WORKED EXAMPLE · Discrete continuity

classical

Problem. Around a circle sit $2n$ people (n even), n wearing red hats and n blue. Show that some n consecutive people include exactly $n/2$ red hats.

Solution. Let r_i be the number of reds in the block starting at seat i . Moving the block by one seat removes one person and adds one, so $|r_{i+1} - r_i| \leq 1$. The block starting at i and the block starting at $i + n$ are complementary, so $r_i + r_{i+n} = n$. Hence the values r_i and $n - r_i$ are both attained; one is $\leq n/2 \leq$ the other, and by discrete continuity $n/2$ is attained in between.

Lesson. a quantity moving in unit steps between a value below and a value above a target must hit it.

■ Common proof shapes

goal	typical skeleton
“for all n ”	induction; or direct construction; or invariant
“exists”	construction; pigeonhole; extremal; probabilistic; IVT
“no solution”	mod m ; size bounds; descent; parity; invariant
“at most N ”	double count; injection into a set of size N ; linear algebra; weight argument
“at least N ”	construction plus averaging; greedy algorithm; Hall
“find all f ”	plug values $\rightarrow$ structure $\rightarrow$ candidate $\rightarrow$ verify
“minimum is m ”	bound + construction (both!)
“process ends”	monovariant in a well-ordered set
“unique”	two solutions $\Rightarrow$ their difference/ratio is trivial

WORKED EXAMPLE · Invariant: a multiplicative one

folklore

Problem. The numbers $1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{100}$ are on a board. A move erases a, b and writes $a + b + ab$. After 99 moves one number remains. Find it.

Solution. $a + b + ab = (1 + a)(1 + b) - 1$, so $\prod(1 + x)$ over the board is invariant. Initially $\prod_{k=1}^{100} (1 + \frac{1}{k}) = \prod \frac{k+1}{k} = 101$. So the last number x satisfies $1 + x = 101$: $x = 100$.

Lesson. if an operation looks like a “shifted” sum or product, change variables ($y = 1 + x$) to make it one.

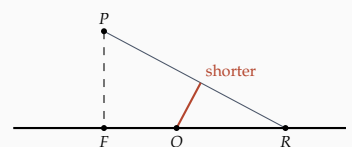
WORKED EXAMPLE · Extremal principle

Sylvester–Gallai

Problem. $n \geq 3$ points in the plane, not all collinear. Show some line passes through exactly two of them.

Solution. Among all pairs (point P , line ℓ through ≥ 2 of the points, $P \notin \ell$) pick one with the distance $d(P, \ell)$ minimal (finitely many pairs, and at least one exists). Let F be the foot from P to ℓ . If ℓ contained three points, two of them, say Q, R , lie on the same side of F (possibly $Q = F$), with Q between F and R . Then $d(Q, PR) < d(P, \ell)$, contradiction. So ℓ has exactly two points.

Lesson. choose the extremal object among a cleverly chosen family, not just among the points.



TRAINING METHODOLOGY

- **Work at the edge.** Choose problems you solve about half the time. Too easy teaches speed only; too hard teaches only reading.
- **Time box.** 45–90 minutes before hints on IMO-level problems; for Putnam A/B1–3 practise 20–30-minute solves.
- **Post-mortem every problem.** Write one line: *what was the key idea, and what in the statement should have signalled it?* This builds the signal→tool map in this booklet.
- **Rewrite solutions** in full after reading an official one. Writing is a separate skill and it is graded.
- **Mixed practice.** Topic-blocked practice builds tools; mixed sets build recognition. Do both.
- **Mock exams** under real timing, including the write-up.
- **Keep a lemma file.** Every reusable configuration or trick, stated precisely, with one problem where it was used.

II ALGEBRA

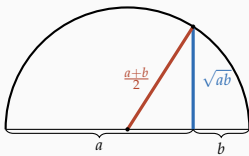
inequalities · polynomials · functional equations · sequences

CLASSICAL INEQUALITIES

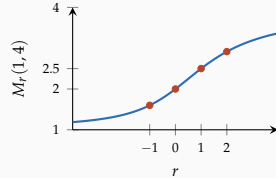
Power means

For $x_i > 0$, weights $w_i > 0$, $\sum w_i = 1$: $M_r = (\sum w_i x_i^r)^{1/r}$, $M_0 = \prod x_i^{w_i}$. Then $r < s \Rightarrow M_r \leq M_s$, equality iff all x_i equal. In particular

$$\sqrt{\frac{\sum x_i^2}{n}} \geq \frac{\sum x_i}{n} \geq \sqrt[n]{\prod x_i} \geq \frac{n}{\sum 1/x_i} \quad (\text{QM} \geq \text{AM} \geq \text{GM} \geq \text{HM}).$$



Left: $\text{GM} \leq \text{AM} \leq \text{radius}$. Right: M_r increases in r (HM, GM, AM, QM marked).



Cauchy–Schwarz & Engel form (Titu)

$(\sum a_i b_i)^2 \leq (\sum a_i^2)(\sum b_i^2)$, equality iff $a \parallel b$.

For $b_i > 0$: $\sum \frac{a_i^2}{b_i} \geq \frac{(\sum a_i)^2}{\sum b_i}$.

Engel trick: to lower-bound $\sum \frac{x}{y+z}$, write $\frac{x}{y+z} = \frac{x^2}{x(y+z)}$ and apply Titu: $\geq \frac{(x+y+z)^2}{2(xy+yz+zx)} \geq \frac{3}{2}$ (Nesbitt). Choosing *what to square* is the whole art.

Hölder

For $a_{ij} \geq 0$, $\lambda_j > 0$, $\sum_j \lambda_j = 1$: $\prod_j (\sum_i a_{ij})^{\lambda_j} \geq \sum_i \prod_j a_{ij}^{\lambda_j}$.

Most used: $(\sum a_i)(\sum b_i)(\sum c_i) \geq (\sum \sqrt[3]{a_i b_i c_i})^3$ and $(\sum \frac{a_i^3}{b_i})(\sum b_i)(\sum 1) \geq (\sum a_i)^3$.

► **SIGNAL:** cube roots, $\sum \frac{a}{\sqrt{b^2+8ac}}$ -type expressions (IMO 2001/2): Hölder with exponents chosen from the equality case.

Jensen & weighted AM-GM

f convex on I , $w_i \geq 0$, $\sum w_i = 1$: $f(\sum w_i x_i) \leq \sum w_i f(x_i)$. With $f = -\ln$: $\sum w_i x_i \geq \prod x_i^{w_i}$ (weighted AM-GM). Young: $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$, $\frac{1}{p} + \frac{1}{q} = 1$.

Rearrangement & Chebyshev

If $a_1 \leq \dots \leq a_n$, $b_1 \leq \dots \leq b_n$ and σ a permutation: $\sum a_i b_{\sigma(i)+1} \leq \sum a_i b_{\sigma(i)}$. Chebyshev: similarly ordered $\Rightarrow n \sum a_i b_i \geq (\sum a_i)(\sum b_i)$.

Schur

$a, b, c \geq 0$, $t > 0$: $a^t(a-b)(a-c) + b^t(b-a)(b-c) + c^t(c-a)(c-b) \geq 0$. $t = 1$: $a^3 + 3abc \geq \sum_{\text{sym}} a^2 b \Leftrightarrow p^3 - 4pq + 9r \geq 0$. Equality: $a = b = c$ or two equal and third 0 (the “Schur signature”).

Muirhead & Karamata

If $(\alpha) \succ (\beta)$ (majorises: sorted partial sums $\geq$, equal totals), then $\sum_{\text{sym}} x^\alpha \geq \sum_{\text{sym}} x^\beta$ for $x_i \geq 0$. Karamata: $(x) \succ (y)$, f convex $\Rightarrow \sum f(x_i) \geq \sum f(y_i)$.

POLYNOMIALS

Vieta & Newton

$P(x) = a_n \prod (x - r_i)$: $e_k = \sum_{i_1 < \dots < i_k} r_{i_1} \dots r_{i_k} = (-1)^k \frac{a_{n-k}}{a_n}$. Power sums $p_k = \sum r_i^k$: $p_k - e_1 p_{k-1} + e_2 p_{k-2} - \dots + (-1)^{k-1} e_{k-1} p_1 + (-1)^k k e_k = 0$ ($k \leq n$); for $k > n$ the recurrence $p_k = e_1 p_{k-1} - \dots + (-1)^{n-1} e_n p_{k-n}$.

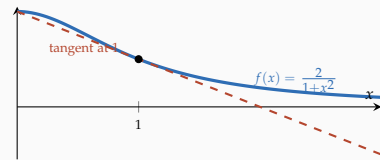
Muirhead needs symmetric sums (all $n!$ terms) and nonnegative variables. Cyclic sums are not symmetric.

Bernoulli

$(1+x)^r \geq 1+rx$ for $x \geq -1$ and $r \geq 1$ or $r \leq 0$; reversed for $0 \leq r \leq 1$.

■ Technique: tangent line trick

If $\sum x_i = n$ is fixed and you need $\sum f(x_i) \geq n f(1)$, try proving $f(x) \geq f(1) + f'(1)(x-1)$ on the domain; summing kills the linear term.



f is not convex, yet $f(x) - (2-x) = \frac{x(x-1)^2}{1+x^2} \geq 0$ on $[0, \infty)$, so $\sum x_i = n \Rightarrow \sum f(x_i) \geq n$. Always verify the sign by factoring out $(x-1)^2$.

■ Technique: SOS (sum of squares)

Write $F(a, b, c) = S_a(b-c)^2 + S_b(c-a)^2 + S_c(a-b)^2$. Then $F \geq 0$ if

- $S_a, S_b, S_c \geq 0$; or
- $a \geq b \geq c$, $S_b \geq 0$, $S_a + S_b \geq 0$, $S_c + S_b \geq 0$ (uses $(a-c)^2 \geq (a-b)^2 + (b-c)^2$).

Useful: $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2} \sum (a-b)^2$, $a^3 + b^3 + c^3 - 3abc = \frac{1}{2} (a+b+c) \sum (a-b)^2$.

■ Technique: pqr / uvw

Let $p = \sum a$, $q = \sum ab$, $r = abc$. Any symmetric polynomial is a polynomial in p, q, r .

uvw principle

Fix p, q (reals $a, b, c \geq 0$). The attainable r form an interval whose endpoints occur when two variables are equal or one is 0. So if the target is monotone, linear, or concave in r (for a minimum), it suffices to check $b = c$ and $c = 0$.

Bounds: $p^2 \geq 3q$, $q^2 \geq 3pr$, $pq \geq 9r$, $p^3 \geq 27r$, Schur $r \geq \frac{p(4q-p^2)}{9}$.

■ Standard substitutions

condition	substitute
$abc = 1$	$a = \frac{x}{y}, b = \frac{y}{z}, c = \frac{z}{x}$ or $a = \frac{x^2}{yz}$
a, b, c sides of triangle	Ravi: $a = y + z$, $b = z + x$, $c = x + y$, $x, y, z > 0$
$a + b + c = abc$	$a = \tan A, \dots$ (acute triangle)
$ab + bc + ca = 1$	$a = \tan \frac{A}{2}, \dots$ or $a = \cot A$
$a^2 + b^2 + c^2 + 2abc = 1$	$a = \cos A, \dots$ (acute)
$a^2 + b^2 + c^2 + abc = 4$	$a = 2 \cos A, \dots$
homogeneous	normalise: $\sum a = 1$ or $abc = 1$ or one var = 1

■ Choosing an inequality

- Find the equality case. If $a = b = c$: AM-GM/C-S/Jensen/SOS. If $(1, 1, 0)$ or $(t, t, 0)$: Schur-type or uvw boundary.
- Homogenise using the constraint, then expand only if degree ≤ 6 and 3 variables.
- Denominators: Titu, or clear them and SOS; or tangent-line on each fraction.
- Square roots: C-S / Hölder / Jensen with concave $\sqrt{\cdot}$, or “SOS-Schur” after substitution.

Integer/rational coefficients

(i) $P \in \mathbb{Z}[x] \Rightarrow a-b \mid P(a) - P(b)$. (ii) Rational root $\frac{p}{q}$ (lowest terms) of $\sum a_i x^i$: $p \mid a_0$, $q \mid a_n$. (iii) **Gauss**: a primitive $P \in \mathbb{Z}[x]$ reducible over $\mathbb{Q}$ is reducible over $\mathbb{Z}$. (iv) **Eisenstein**: $p \mid a_i$ ($i < n$), $p \nmid a_n$, $p^2 \nmid a_0 \Rightarrow$ irreducible. Try $x \mapsto x+1$: $\Phi_p(x) = \frac{x^p-1}{x-1}$ irreducible.

Interpolation & finite differences

Lagrange: $P(x) = \sum_i y_i \prod_{j \neq i} \frac{x-x_j}{x_i-x_j}$ is the unique poly of degree $< n$ through n points.
 $\Delta P(x) = P(x+1) - P(x)$ lowers degree by 1; for $\deg P < n$:
 $\sum_{k=0}^n (-1)^k \binom{n}{k} P(k) = 0$.
 P is integer-valued on $\mathbb{Z} \iff P(x) = \sum c_k \binom{x}{k}, c_k \in \mathbb{Z}$.

" $P(k) = \frac{1}{k+1}$ for $k = 0..n$, find $P(n+1)$ ": set $Q(x) = (x+1)P(x) - 1$, which has roots $0, \dots, n$, so $Q(x) = c x(x-1) \cdots (x-n)$; find c from $x = -1$.

Roots-of-unity filter

$\omega = e^{2\pi i/m}$: $\sum_{k \equiv r(m)} a_k = \frac{1}{m} \sum_{j=0}^{m-1} \omega^{-jr} F(\omega^j)$ for $F(x) = \sum a_k x^k$.
 Ex: $\sum_k \binom{n}{3k} = \frac{1}{3} (2^n + 2 \cos \frac{n\pi}{3})$.

Chebyshev polynomials

$T_n(\cos \theta) = \cos n\theta$, $T_{n+1} = 2xT_n - T_{n-1}$, leading coeff 2^{n-1} . Any monic P of degree n has $\max_{[-1,1]} |P| \geq 2^{1-n}$, equality for $2^{1-n} T_n$.

Other tools. $P(x) - x \mid P(P(x)) - x$. Complex roots of real polys pair up. Descartes' rule of signs bounds positive roots. Root bound $|z| < 1 + \max_{i < n} |a_i/a_n|$. Mason–Stothers: coprime $a + b = c$ in $\mathbb{C}[x]$, not all constant $\Rightarrow \max \deg \leq n_0(abc) - 1$ (distinct roots); this kills $f^n + g^n = h^n$ for $n \geq 3$.

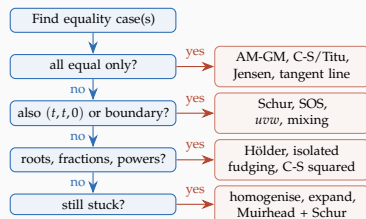
" $P(x) = Q(x)$ for infinitely many x " gives $P \equiv Q$; "for all $x \in \mathbb{Z}$ mod p " does not ($x^p \equiv x$).

FUNCTIONAL EQUATIONS

Write $P(x, y)$ for the assertion; record every substitution.

- Plug in** $0, 1, -1, x = y, y = -x, y = f(x), y = \frac{1}{x}$, values that zero out a term.
- Injectivity:** $f(a) = f(b) \Rightarrow$ plug to get $a = b$. **Surjectivity:** exhibit $f(\text{something}) = \text{any } t$. Look for $f(\text{expr}) = \text{linear in } x$.
- Symmetry swap:** compare $P(x, y)$ and $P(y, x)$.
- Fixed points / involutions:** $f(f(x)) = x$ makes f bijective.
- Guess the answer** (usually $f \equiv 0, x, cx, x^2, 1-x, \frac{1}{x}$) and set $g = f - \text{answer}$.
- Over $\mathbb{N}$: induction, growth rates, minimality, parity of f .

MORE INEQUALITY TECHNIQUES



A first-pass decision path. The equality case is the most informative single fact about an inequality.

Multilinear & convex-in-each-variable

If $F(x_1, \dots, x_n)$ is linear (or convex) in each variable separately on a box $\prod [a_i, b_i]$, then $\max F$ is attained at a vertex ($x_i \in \{a_i, b_i\}$).
 ► **SIGNAL:** " $0 \leq x_i \leq 1$, maximise $\sum x_i - \sum x_i x_{i+1}$ ".

Isolated fudging

To show $\sum_{\text{cyc}} f(a, b, c) \geq 1$, find r with $f(a, b, c) \geq \frac{a^r}{a^r + b^r + c^r}$ for every term; the right sides sum to 1. Determine r by matching the tangent at the equality case.

Cauchy's equation $f(x+y) = f(x) + f(y)$

On $\mathbb{Q}$: $f(x) = cx$. On $\mathbb{R}$, $f(x) = cx$ if additionally any one of: continuous at a point; monotone; bounded above/below on some interval; $f(x) \geq 0$ for $x \geq 0$; measurable. Otherwise wild (Hamel basis) solutions exist.

equation (nice f)	solutions
$f(x+y) = f(x)f(y)$	$0, e^{cx}$
$f(xy) = f(x) + f(y), x > 0$	$c \ln x$
$f(xy) = f(x)f(y), x > 0$	$0, 1, x^c$
$f(\frac{x+y}{2}) = \frac{f(x)+f(y)}{2}$ (Jensen)	$ax + b$
$f(x+y) + f(x-y) = 2f(x) + 2f(y)$	cx^2
$f(x+y) + f(x-y) = 2f(x)f(y)$	$\cos cx, \cosh cx, 0$

Finding $f(x)^2 = x^2$ for all x does *not* give $f(x) = x$ or $f(x) = -x$ globally; handle the *pointwise trap*: assume $f(a) = a, f(b) = -b$ with $a, b \neq 0$ and derive a contradiction.

SEQUENCES & RECURRENCES

Linear recurrences

$a_{n+k} = c_1 a_{n+k-1} + \dots + c_k a_n$: roots λ of $x^k - c_1 x^{k-1} - \dots - c_k$; root of multiplicity m gives $n^j \lambda^n, j < m$. Inhomogeneous: add a particular solution.

Nonlinear recurrences. Look for invariants: Ex. $a_{n+1} = \frac{a_n^2 + c}{a_{n-1}}$ gives $\frac{a_{n+1} + a_{n-1}}{a_n}$ constant, so the sequence is *linear* and integrality follows.
 $a_{n+1} = a_n^2 - 2$: substitute $a_n = t^{2^n} + t^{-2^n}$. $a_{n+1} = 2a_n^2 - 1$: $\cos(2^n \theta)$ or $\cosh$.

Fibonacci facts

$F_{m+n} = F_m F_{n+1} + F_{m-1} F_n$; $\gcd(F_m, F_n) = F_{\gcd(m, n)}$; Cassini $F_{n+1} F_{n-1} - F_n^2 = (-1)^n$; $F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$; Zeckendorf: unique sum of non-consecutive Fibonacci numbers; $\sum_{k \leq n} F_k = F_{n+2} - 1$.

Telescoping. $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$; $\frac{1}{k(k+1)(k+2)} = \frac{1}{2} (\frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)})$; $\arctan \frac{1}{k^2+k+1} = \arctan(k+1) - \arctan k$; $k \cdot k! = (k+1)! - k!$. **Abel summation:** $\sum_{i=1}^n a_i b_i = A_n b_n + \sum_{i=1}^{n-1} A_i (b_i - b_{i+1})$, $A_i = \sum_{j \leq i} a_j$.

WORKED EXAMPLE · Isolated fudging

IMO 2001/2

Problem. $a, b, c > 0$. Prove $\sum_{\text{cyc}} \frac{a}{\sqrt{a^2 + 8bc}} \geq 1$.

Solution. Claim: $\frac{a}{\sqrt{a^2 + 8bc}} \geq \frac{a^{4/3}}{a^{4/3} + b^{4/3} + c^{4/3}}$. Squaring, this is $(a^{4/3} + b^{4/3} + c^{4/3})^2 \geq a^{2/3}(a^2 + 8bc)$, i.e.

$$(b^{4/3} + c^{4/3})(2a^{4/3} + b^{4/3} + c^{4/3}) \geq 8a^{2/3}bc.$$

By AM-GM, $b^{4/3} + c^{4/3} \geq 2b^{2/3}c^{2/3}$ and $2a^{4/3} + b^{4/3} + c^{4/3} \geq 4a^{2/3}b^{1/3}c^{1/3}$; multiply. Summing the claim over the three terms gives ≥ 1 .

Lesson. when each term should be compared to a "share" of 1, guess $a^r / \sum a^r$ and solve for r .

WORKED EXAMPLE · Substitution + Titu

IMO 1995/2

Problem. $a, b, c > 0, abc = 1$. Prove $\sum \frac{1}{a^3(b+c)} \geq \frac{3}{2}$.

Solution. Put $x = \frac{1}{a}, y = \frac{1}{b}, z = \frac{1}{c}$, so $xyz = 1$ and $\frac{1}{a^3(b+c)} = \frac{x^3}{\frac{1}{y} + \frac{1}{z}} = \frac{x^3 yz}{y+z} = \frac{x^2}{y+z}$. By Titu,

$$\sum \frac{x^2}{y+z} \geq \frac{(x+y+z)^2}{2(x+y+z)} = \frac{x+y+z}{2} \geq \frac{3\sqrt[3]{xyz}}{2} = \frac{3}{2}.$$

Lesson. an ugly denominator involving $abc = 1$ often simplifies under reciprocals.

WORKED EXAMPLE · Tangent line trick

Poland 1996

Problem. $a + b + c = 1$, $a, b, c \geq -\frac{3}{4}$. Prove $\sum \frac{a}{a^2 + 1} \leq \frac{9}{10}$.

Solution. Equality at $a = b = c = \frac{1}{3}$. The tangent to $f(x) = \frac{x}{x^2 + 1}$ at $\frac{1}{3}$ is $\frac{18x}{25} + \frac{3}{50}$, and

$$\frac{18x}{25} + \frac{3}{50} - \frac{x}{x^2 + 1} = \frac{(3x-1)^2(4x+3)}{50(x^2+1)} \geq 0 \quad (x \geq -\frac{3}{4}).$$

Summing: $\sum f(a) \leq \frac{18}{25} + \frac{9}{50} = \frac{9}{10}$.

Lesson. the odd-looking constraint $-\frac{3}{4}$ is exactly where the factor $4x+3$ changes sign; constraints are clues.

WORKED EXAMPLE · pqr with Schur

classical

Problem. $a, b, c \geq 0$, $a + b + c = 3$. Prove $a^2 + b^2 + c^2 + abc \geq 4$.

Solution. With $p = 3$: $a^2 + b^2 + c^2 = 9 - 2q$, so we need $r \geq 2q - 5$.

Schur gives $r \geq \frac{p(4q-p^2)}{9} = \frac{4q-9}{3}$, and $\frac{4q-9}{3} \geq 2q-5 \iff q \leq 3$, true as $q \leq \frac{p^2}{3}$. Equality at $a = b = c = 1$.

Lesson. fix p , translate to q, r ; Schur is the standard lower bound for r .

Lagrange multipliers (fine on the Putnam; risky at the IMO unless you handle the boundary and existence of a maximum rigorously). At an interior extremum of f subject to $g = c$: $\nabla f = \lambda \nabla g$. Always also check the boundary and behaviour at infinity.

SOS with a twist. For cyclic (not symmetric) inequalities, write the difference as $\sum S_a(b-c)^2$ with S_a depending on all three variables, then use the criteria of §Algebra. For degree 4 cyclic, the standard " $a^3b + b^3c + c^3a \leq \frac{(a^2+b^2+c^2)^2}{3}$ " (Vasc) is worth remembering.

MORE POLYNOMIAL TOOLS

Combinatorial Nullstellensatz

Let $f \in F[x_1, \dots, x_n]$ have degree $\sum t_i$ with the coefficient of $\prod x_i^{t_i}$ nonzero. If $S_i \subseteq F$, $|S_i| > t_i$, then some $s \in \prod S_i$ has $f(s) \neq 0$.

Cauchy–Davenport. $A, B \subseteq \mathbb{Z}/p$ nonempty $\Rightarrow |A+B| \geq \min(p, |A|+|B|-1)$.

Integer polynomials under iteration

$P \in \mathbb{Z}[x]$: $a-b \mid P(a)-P(b)$, so along an orbit $a \rightarrow P(a) \rightarrow \dots$ the consecutive differences divide each other. Consequence: every integer periodic point of P has period 1 or 2. (IMO 2006/5: $P^{\circ k}(x) = x$ has at most $\deg P$ integer solutions for $\deg P \geq 2$.)

Toolkit for "find all polynomials". Compare degrees and leading coefficients; plug in roots; if $P(x) \mid P(g(x))$ -type relations hold, study how roots move under g (a root r forces $g(r)$ to be a root; infinitely many roots $\Rightarrow P \equiv 0$); consider $P(x) - x$, $P(x) - P(0)$; use $a-b \mid P(a)-P(b)$ with huge a .

WORKED EXAMPLE · Interpolation

USAMO 1975

Problem. $\deg P = n$ and $P(k) = \frac{k}{k+1}$ for $k = 0, 1, \dots, n$. Find $P(n+1)$.

Solution. $Q(x) = (x+1)P(x) - x$ has degree $n+1$ and roots $0, \dots, n$: $Q(x) = c x(x-1)\dots(x-n)$. At $x = -1$: $1 = c(-1)^{n+1}(n+1)!$. Then $(n+2)P(n+1) - (n+1) = Q(n+1) = c(n+1)! = (-1)^{n+1}$, so

$$P(n+1) = \frac{n+1+(-1)^{n+1}}{n+2}.$$

Lesson. multiply away the denominator to create a polynomial with known roots.

WORKED EXAMPLE · Roots of unity

classical

Problem. For which n does $x^2 + x + 1$ divide $x^{2n} + x^n + 1$?

Solution. The roots of $x^2 + x + 1$ are $\omega, \bar{\omega}$, primitive cube roots of 1. If $3 \nmid n$, ω^n is again a primitive cube root, so $\omega^{2n} + \omega^n + 1 = 0$ (same for $\bar{\omega}$): divisible. If $3 \mid n$, the value is $3 \neq 0$. Answer: $3 \nmid n$.

Lesson. divisibility by a polynomial with known roots = vanishing at those roots.

MORE FUNCTIONAL-EQUATION TECHNIQUE

- Get $f(0)$, then f at special points.** Many FEs collapse once $f(0)$ is known.
- Surjectivity from a free variable:** if $f(\text{expr}(x)) = \text{linear in } x$, f is onto.
- Injectivity:** from $f(a) = f(b)$ plug to get something like $a + c = b + c$.
- Kill terms:** choose y so that the inside of one f equals the inside of another.
- Iterate:** apply f to both sides, or substitute the equation into itself.
- Additive structure:** derive $f(x+y) = f(x) + f(y) - f(0)$ and invoke Cauchy with a regularity condition you have proven (Ex. f bounded below on an interval).
- Over $\mathbb{N}$ or $\mathbb{Z}$:** get a recurrence $f(n+1) = f(n) + c$ and solve.

WORKED EXAMPLE · Linearisation over $\mathbb{Z}$

IMO 2019/1

Problem. Find all $f: \mathbb{Z} \rightarrow \mathbb{Z}$ with $f(2a) + 2f(b) = f(f(a+b))$.

Solution. $a = 0$: $f(0) + 2f(b) = f(f(b))$. Replacing b by $a+b$: $f(f(a+b)) = f(0) + 2f(a+b)$. Hence $f(2a) + 2f(b) = f(0) + 2f(a+b)$. With $a = 1$: $f(b+1) - f(b) = \frac{f(2)-f(0)}{2}$, a constant, so $f(x) = mx + c$. Substituting: $2ma + c + 2mb + 2c = m^2(a+b) + mc + c$, so $m^2 = 2m$ and $2c = mc$. Answer: $f \equiv 0$, or $f(x) = 2x + c$ for any $c \in \mathbb{Z}$ (both check).

Lesson. use one substitution to express the "hard" side twice; subtract.

WORKED EXAMPLE · Surjectivity

ISL 2002 A1

Problem. Find all $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(f(x) + y) = 2x + f(f(y) - x)$.

Solution. $y = -f(x)$: $f(0) = 2x + f(f(-f(x)) - x)$, so f takes the value $f(0) - 2x$ for every x : f is surjective. Pick a with $f(a) = 0$. Put $x = a$: $f(y) = 2a + f(f(y) - a)$. Since $f(y)$ ranges over all $z \in \mathbb{R}$: $f(z-a) = z - 2a$, i.e. $f(t) = t - a$. So $f(x) = x + c$; these work.

Lesson. a free variable appearing linearly outside f gives surjectivity; then use a zero of f .

Always verify the candidate in the original equation, and state it. Missing verification is the most common lost point in FEs.

SEQUENCES IN ACTION

WORKED EXAMPLE · Integrality via an invariant

classical

Problem. $a_1 = a_2 = 1$, $a_{n+2} = \frac{a_{n+1}^2 + 2}{a_n}$. Prove all a_n are integers.

Solution. $a_{n+2}a_n - a_{n+1}^2 = 2$ for all n . Subtracting consecutive instances: $a_{n+2}a_n - a_{n+1}^2 = a_{n+1}a_{n-1} - a_n^2$, i.e. $a_n(a_{n+2} + a_n) = a_{n+1}(a_{n+1} + a_{n-1})$. So $\frac{a_{n+2} + a_n}{a_{n+1}}$ is constant $= \frac{a_3 + a_1}{a_2} = 4$. Hence $a_{n+2} = 4a_{n+1} - a_n$: integers.

Lesson. a "determinant" $a_{n+2}a_n - a_{n+1}^2$ constant $\Rightarrow$ a linear recurrence exists.

Bounding sequences. Squeeze a_n between two sequences you can solve; compare to $\sqrt{n}$, $\log n$ via $a_{n+1}^2 - a_n^2$ or $\frac{1}{a_{n+1}} - \frac{1}{a_n}$ (telescoping differences of a function of a_n). Ex. $a_{n+1} = a_n + \frac{1}{a_n}$: $a_{n+1}^2 = a_n^2 + 2 + \frac{1}{a_n^2}$, so $a_n \sim \sqrt{2n}$.

III NUMBER THEORY

divisibility · orders · valuations · residues · Diophantine equations

FOUNDATIONS

Bézout, Euclid, CRT

$\gcd(a, b) = \min\{ax + by > 0\}$; $\gcd(a, b) = 1 \iff \exists x, y : ax + by = 1$. Euclid's lemma: $p \mid ab \Rightarrow p \mid a$ or $p \mid b$. $\gcd(a^m - 1, a^n - 1) = a^{\gcd(m, n)} - 1$. **CRT**: pairwise coprime $m_i \Rightarrow x \equiv a_i \pmod{m_i}$ has a unique solution mod $\prod m_i$.

Fermat, Euler, Wilson

$\gcd(a, n) = 1 \Rightarrow a^{\varphi(n)} \equiv 1 \pmod{n}$; $a^p \equiv a \pmod{p}$. Wilson: $(p-1)! \equiv -1 \pmod{p}$, and $n > 4$ composite $\Rightarrow (n-1)! \equiv 0 \pmod{n}$. $\varphi(n) = n \prod_{p \mid n} (1 - \frac{1}{p})$, $\sum_{d \mid n} \varphi(d) = n$.

Multiplicative functions

$\tau(n) = \prod (e_i + 1)$, $\sigma(n) = \prod \frac{p_i^{e_i+1} - 1}{p_i - 1}$. $\tau(n)$ odd $\iff n$ square. Möbius inversion: $g(n) = \sum_{d \mid n} f(d) \iff f(n) = \sum_{d \mid n} \mu(d) g(n/d)$.

ORDERS & PRIMITIVE ROOTS

Order

$\text{ord}_n(a) = \text{least } k > 0 \text{ with } a^k \equiv 1 \pmod{n}$. Then $a^m \equiv 1 \iff \text{ord}_n(a) \mid m$. Hence $\text{ord}_n(a) \mid \varphi(n)$.

The order argument (the single most important NT move). If $p \mid a^m - 1$ and $p \nmid a^{m/q} - 1$ for each prime $q \mid m$, then $\text{ord}_p(a) = m$, so $m \mid p-1$. Ex. every prime divisor p of $2^{2^n} + 1$ satisfies $p \equiv 1 \pmod{2^{n+1}}$ (in fact mod 2^{n+2} for $n \geq 2$).

Primitive roots

A generator of $(\mathbb{Z}/n)^\times$ exists iff $n = 1, 2, 4, p^k, 2p^k$ (p odd). For p prime, $(\mathbb{Z}/p)^\times$ is cyclic of order $p-1$; $x^d \equiv 1$ has exactly $\gcd(d, p-1)$ solutions.

Sums of powers mod p : $\sum_{x=1}^{p-1} x^k \equiv \begin{cases} -1 & (p-1) \mid k \\ 0 & \text{else} \end{cases} \pmod{p}$. Also: Wolstenholme, $p \geq 5$: $\sum_{k=1}^{p-1} \frac{1}{k} \equiv 0 \pmod{p^2}$.

p-ADIC VALUATIONS

$v_p(n) = \text{exponent of } p \text{ in } n$. $v_p(ab) = v_p(a) + v_p(b)$; $v_p(a+b) \geq \min$, with equality if the two differ.

Legendre & Kummer

$v_p(n!) = \sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor = \frac{n - s_p(n)}{p-1}$ (s_p = base- p digit sum).

Kummer: $v_p\binom{m+n}{m} = \text{number of carries when adding } m, n \text{ in base } p$. Lucas: $\binom{m}{n} \equiv \prod \binom{m_i}{n_i} \pmod{p}$ over base- p digits.

Lifting the Exponent (LTE)

p odd prime, $p \mid a-b$, $p \nmid a, b$: $v_p(a^n - b^n) = v_p(a-b) + v_p(n)$.
 n odd, $p \mid a+b$: $v_p(a^n + b^n) = v_p(a+b) + v_p(n)$.
 $p = 2$, a, b odd, n even: $v_2(a^n - b^n) = v_2(a-b) + v_2(a+b) + v_2(n) - 1$.

LTE needs $p \mid a-b$ (not merely $p \mid a^n - b^n$). If only $p \mid a^n - b^n$, first pass to $a^d - b^d$ with $d = \text{ord}_p(a/b)$.

Zsigmondy

For $a > b \geq 1$ coprime, $n \geq 2$, $a^n - b^n$ has a prime divisor not dividing $a^k - b^k$ for any $k < n$, except $(a, b, n) = (2, 1, 6)$ and $n = 2$ with $a+b$ a power of 2. Similarly $a^n + b^n$ ($n \geq 2$) except $2^3 + 1^3$.

► **SIGNAL**: "find all n with $a^n - 1 \mid \dots$ ", " $x^n - 1$ is a prime power". Zsigmondy is heavy; LTE + orders often suffice and are safer to cite.

QUADRATIC RESIDUES

Legendre symbol

$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$ (Euler). Multiplicative. $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$, $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$.

Reciprocity: $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}$ for odd primes $p \neq q$.

a is a QR mod p iff

-1	$p \equiv 1 \pmod{4}$
2	$p \equiv \pm 1 \pmod{8}$
-2	$p \equiv 1, 3 \pmod{8}$
3	$p \equiv \pm 1 \pmod{12}$
-3	$p \equiv 1 \pmod{3}$
5	$p \equiv \pm 1 \pmod{5}$

Consequences. $p \mid x^2 + 1 \Rightarrow p = 2$ or $p \equiv 1 \pmod{4}$. $p \mid x^2 + xy + y^2$ (co-prime) $\Rightarrow p = 3$ or $p \equiv 1 \pmod{3}$. Infinitely many primes $\equiv 1 \pmod{n}$: prime factors of $\Phi_n(kn)$.

Sums of squares

n is a sum of two squares $\iff$ every prime $\equiv 3 \pmod{4}$ appears to an even power. $p \equiv 1 \pmod{4}$ is a sum of two squares uniquely. Every n is a sum of four squares; n is a sum of three $\iff n \neq 4^a(8b+7)$.

DIOPHANTINE EQUATIONS

- Mod out**. Squares mod 3,4,5,8: $\{0,1\}$, $\{0,1\}$, $\{0,\pm 1\}$, $\{0,1,4\}$. Cubes mod 7, 9: $\{0,\pm 1\}$. Fourth powers mod 16: $\{0,1\}$. Pick the modulus where the powers have few residues (m with $\varphi(m)$ divisible by the exponent).
- Factor**. $x^2 - y^2 = (x-y)(x+y)$; Simon's trick $xy + ax + by = c \iff (x+b)(y+a) = c + ab$; sum/difference of cubes; in $\mathbb{Z}[i]$ or $\mathbb{Z}[\omega]$.
- Bound**. Squeeze between consecutive powers: $(x^2)^2 < x^4 + x + 1 < (x^2 + 1)^2$ for $x \geq 1$, so $x^4 + x + 1$ is never a square.
- gcd analysis**. If $ab = c^2$, $\gcd(a, b) = 1$, $a, b > 0$ then both are squares.
- Descent / Vieta jumping** (below).
- Size of valuations**: compare v_p of both sides (LTE).

Pythagorean triples & Pell

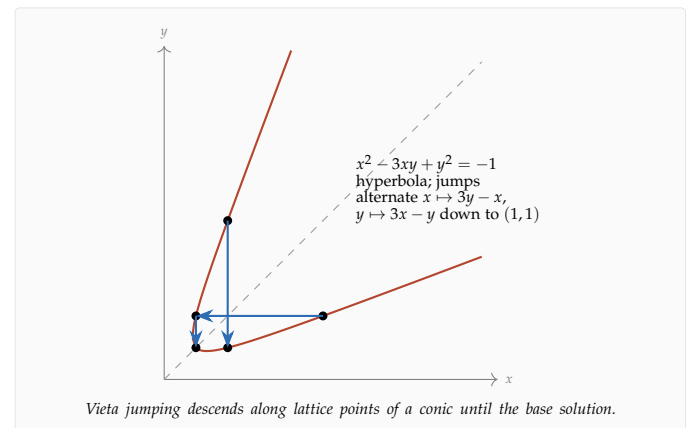
Primitive $x^2 + y^2 = z^2$: $(m^2 - n^2, 2mn, m^2 + n^2)$, $\gcd(m, n) = 1$, $m \not\equiv n \pmod{2}$.

Pell $x^2 - dy^2 = 1$ (d non-square): all solutions $x + y\sqrt{d} = \pm(x_1 + y_1\sqrt{d})^k$ from the fundamental one; x_k, y_k satisfy $u_{k+1} = 2x_1u_k - u_{k-1}$.

Vieta jumping (root flipping)

Given $x^2 - kxy + y^2 = c$ -type equations symmetric in x, y : fix a solution (x, y) with $x + y$ minimal, $x \geq y$. Treat as quadratic in x ; the other root $x' = ky - x = \frac{y^2 - c}{x}$ is an integer. Show $0 < x' < x$ (or $x' = 0$), contradicting minimality or forcing the base case.

IMO 1988/6: $ab + 1 \mid a^2 + b^2 \Rightarrow \frac{a^2 + b^2}{ab + 1}$ is a perfect square.



Vieta jumping descends along lattice points of a conic until the base solution.

Useful identities. Sophie Germain $a^4 + 4b^4 = (a^2 + 2b^2 + 2ab)(a^2 + 2b^2 - 2ab)$. Brahmagupta $(a^2 + nb^2)(c^2 + nd^2) = (ac \mp nbd)^2 + n(ad \pm b^2c)^2$. $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$. $x^n - y^n = (x - y) \sum x^{n-1-k} y^k$.

MISCELLANY THAT SHOWS UP

- **Cyclotomic:** $x^n - 1 = \prod_{d|n} \Phi_d(x)$; prime $p \mid \Phi_n(a) \Rightarrow p \mid n$ or $p \equiv 1 \pmod{n}$.
- **Floor sums:** $\lfloor x \rfloor + \lfloor x + \frac{1}{n} \rfloor + \cdots + \lfloor x + \frac{n-1}{n} \rfloor = \lfloor nx \rfloor$ (Hermite).
 $\sum_{k=1}^{q-1} \lfloor \frac{kp}{q} \rfloor = \frac{(p-1)(q-1)}{2}$ for coprime p, q .
- **Beatty:** $\frac{1}{\alpha} + \frac{1}{\beta} = 1$, irrational $\alpha, \beta > 1 \Rightarrow \{\lfloor n\alpha \rfloor\}, \{\lfloor n\beta \rfloor\}$ partition $\mathbb{N}$.
- **Chicken McNugget:** coprime a, b : largest non-representable $ab - a - b$; exactly $\frac{(a-1)(b-1)}{2}$ non-representable.
- **Dirichlet approximation:** $|\alpha - \frac{p}{q}| < \frac{1}{q^2}$ infinitely often for irrational

MORE NUMBER-THEORY TECHNIQUE

Size + divisibility

If $a \mid b$ and $0 < b < 2a$, then $b = a$. If $a \mid b$ with $b \neq 0$ then $|a| \leq |b|$.
► SIGNAL: “ $ab + 1 \mid a^2 + b^2$ ”, “ $n^2 \mid \dots$ ”: combine a divisibility with an inequality to force equality or a small case.

Schur’s theorem on prime divisors

A nonconstant $P \in \mathbb{Z}[x]$ has infinitely many primes dividing some value $P(n)$. *Proof idea:* $P(cp_1 \cdots p_k P(0))/P(0) \equiv 1$ modulo each p_i .

CRT constructions. To build x with several local properties (divisible by a square, avoiding residues, prescribed remainders), impose each property modulo a different prime (power) and glue by CRT. Ex. n consecutive integers each divisible by a square > 1 : choose distinct primes $p_1, \dots, p_n$ and $x \equiv -i \pmod{p_i^2}$.

Look at the smallest/largest prime factor. The smallest prime p of n satisfies $\gcd(n, p-1) = 1$, which interacts beautifully with orders ($\text{ord}_p(a) \mid \gcd(n, p-1)$). The largest prime factor is useful for size bounds and in factorial problems.

Exponent equations $a^x \pm b^y = c^z$: reduce mod small numbers to fix parities; factor as difference of squares when an exponent is even; LTE or Zsigmondy for the rest; bound by size.

Infinite descent. Show a solution produces a strictly smaller positive one. Classical: $x^4 + y^4 = z^2$ has no positive solutions; $\sqrt{2}$ irrational; Vieta jumping is structured descent.

Sequences mod m . A sequence defined by a recurrence in finitely many previous terms is eventually periodic mod m ; purely periodic if the recurrence is reversible (Ex. Fibonacci mod m).

Quadratic reciprocity as a tool. To show $p \mid x^2 - a$ is impossible, compute $(\frac{a}{p}) = -1$. To produce infinitely many primes in a class, find a polynomial whose prime divisors are forced into that class ($x^2 + 1$ gives 1 mod 4; $x^2 + x + 1$ gives 1 mod 3; Φ_n gives 1 mod n).

WORKED EXAMPLE · Orders + LTE

IMO 1990/3

Problem. Find all $n > 1$ with $n^2 \mid 2^n + 1$.

Solution. n is odd. Let p be the smallest prime factor of n . Then $2^{2^n} \equiv 1 \pmod{p}$, so $\text{ord}_p(2) \mid \gcd(2n, p-1) = 2$ (every prime of n is $\geq p$). So $p \mid 2^2 - 1 = 3$: $p = 3$. LTE: $v_3(2^n + 1) = v_3(3) + v_3(n) = 1 + v_3(n) \geq 2v_3(n)$, so $v_3(n) = 1$. If $n \neq 3$, write $n = 3m$, $3 \nmid m$, $m > 1$, and let q be the smallest prime of m . As before $\text{ord}_q(2) \mid \gcd(2n, q-1) \mid 6$, so $q \mid 2^6 - 1 = 63$, $q = 7$. But $2^k \bmod 7 \in \{1, 2, 4\}$, so $7 \nmid 2^n + 1$. Contradiction. Answer: $n = 3$ (and trivially $n = 1$).

Lesson. smallest prime factor + order gives a tiny order; LTE controls the exponent.

α ; $\{n\alpha\}$ dense in $[0, 1)$ (equidistributed).

- **Bertrand:** a prime in $(n, 2n]$ for $n \geq 1$.
- **Thue’s lemma:** $p \nmid a \Rightarrow \exists 0 < |x|, |y| < \sqrt{p}$ with $x \equiv ay \pmod{p}$.
- **Erdős–Ginzburg–Ziv:** any $2n - 1$ integers contain n with sum $\equiv 0 \pmod{n}$.
- **Hensel:** simple root mod p lifts uniquely to mod p^k .
- **Primes in $\mathbb{Z}[i]$:** $1 + i, p \equiv 3 \pmod{4}$, and $a \pm bi$ with $a^2 + b^2 = p \equiv 1 \pmod{4}$.

“ $a \mid b$ and $a \mid c$ ” in problems about $n^2 + 1 \mid \dots$: always check *small* and *negative* cases, and $n = 0$ if allowed.

WORKED EXAMPLE · Vieta jumping

IMO 1988/6

Problem. $a, b \in \mathbb{N}$ and $ab + 1 \mid a^2 + b^2$. Show $k = \frac{a^2 + b^2}{ab + 1}$ is a perfect square.

Solution. Suppose not; among solutions with this k take $a \geq b \geq 1$ with $a + b$ minimal. a is a root of $x^2 - kbx + b^2 - k = 0$; the other root $a' = kb - a = \frac{b^2 - k}{a}$ is an integer. $a' \neq 0$ (else $k = b^2$). If $a' < 0$, then $a'^2 - kba' + b^2 - k \geq a'^2 + k + b^2 - k > 0$, impossible. So $a' > 0$ and $a' = \frac{b^2 - k}{a} < \frac{b^2}{a} \leq a$. Then (a', b) is a solution with smaller sum. Contradiction.

Lesson. the second root of the quadratic is both an integer (Vieta sum) and smaller (Vieta product).

WORKED EXAMPLE · Factor the difference of squares

Catalan’s equation,

small case

Problem. Solve $3^x - 2^y = 1$ in positive integers.

Solution. If $y = 1$: $x = 1$. If $y \geq 2$: $3^x \equiv 1 \pmod{4}$ forces $x = 2m$ even. Then $(3^m - 1)(3^m + 1) = 2^y$; both factors are powers of 2 differing by 2, so they are 2 and 4: $m = 1$, $x = 2$, $y = 3$. Answer: $(1, 1), (2, 3)$.

Lesson. a modulus fixes the parity of an exponent; then factor.

WORKED EXAMPLE · Smallest prime + order

classical

Problem. Find all $n \geq 1$ with $n \mid 2^n - 1$.

Solution. If $n > 1$ let p be its smallest prime; p is odd. $\text{ord}_p(2) \mid n$ and $\text{ord}_p(2) \mid p - 1$, but $\gcd(n, p - 1) = 1$. So $\text{ord}_p(2) = 1$, i.e. $p \mid 1$. Only $n = 1$.

WORKED EXAMPLE · Kummer

classical

Problem. Show $\binom{2n}{n}$ is even for $n \geq 1$, and $v_2(\binom{2n}{n}) = s_2(n)$ (binary digit sum).

Solution. Adding $n + n$ in base 2 shifts n left: a carry occurs exactly at each 1-bit of n . By Kummer, $v_2(\binom{2n}{n})$ is the number of carries $= s_2(n) \geq 1$.

WORKED EXAMPLE · Sums of two squares

classical

Problem. Show that if $p \equiv 1 \pmod{4}$ then $p = x^2 + y^2$.

Solution. -1 is a QR, so $p \mid a^2 + 1$ for some a . By Thue’s lemma there are $0 < |x|, |y| < \sqrt{p}$ with $x \equiv ay \pmod{p}$. Then $x^2 + y^2 \equiv y^2(a^2 + 1) \equiv 0$ and $0 < x^2 + y^2 < 2p$, so $x^2 + y^2 = p$.

Lesson. “small solution of a congruence” (Thue, pigeonhole on a box) + size gives equality.

STANDARD RESULTS WORTH CITING

result	statement
Fermat’s $x^4 + y^4 = z^2$	no positive solutions
Mihăilescu (Catalan)	$x^a - y^b = 1$, $a, b > 1$: only $3^2 - 2^3$
$x^2 + y^2 = z^2$	parametrised (see above)
Markov $x^2 + y^2 + z^2 = 3xyz$	solutions generated from $(1, 1, 1)$ by Vieta jumps
$x^3 + y^3 = z^3$	no positive solutions
Lagrange	every n is a sum of 4 squares
Dirichlet	$\gcd(a, m) = 1 \Rightarrow$ infinitely many primes $\equiv a \pmod{m}$
Chebyshev bound	$\prod_{p \leq n} p < 4^n$

Heavy theorems (Dirichlet, Zsigmondy, Mihăilescu) are allowed to be quoted at the IMO, but if the problem is essentially that theorem, graders expect a proof. When in doubt, prove the special case.

IV COMBINATORICS

counting · existence · processes · graphs · games

EXISTENCE PRINCIPLES

Pigeonhole

n objects in k boxes $\Rightarrow$ some box has $\geq \lceil n/k \rceil$. Infinite version: infinitely many objects, finitely many boxes $\Rightarrow$ one box infinite. Averaging: some term $\geq$ the mean.

Choose the boxes. Residues mod n (prefix sums $S_0, \dots, S_n$: two agree $\Rightarrow$ some block sum $\equiv 0$). Pairs $\{k, 2n+1-k\}$. Odd parts ($m = 2^a \cdot \text{odd}$: among $n+1$ numbers in $[2n]$, one divides another). Sub-squares of a grid.

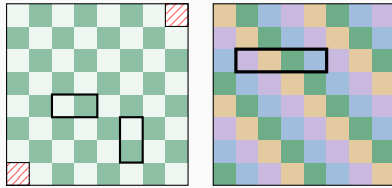
Erdős–Szekeres

Every sequence of $rs+1$ distinct reals has an increasing subsequence of length $r+1$ or a decreasing one of length $s+1$. *Proof:* label each term with (longest inc. ending here, longest dec. ending here); labels are distinct.

Extremal principle. Take the longest path in a graph (endpoint's neighbours lie on the path); the closest pair of points; the smallest counterexample; the vertex of max degree; the convex hull.

INVARIANTS, MONOINVARIANTS, COLOURINGS

- Parity; sum mod m ; product; weighted sums $\sum x^{d(\text{cell})}$ (Conway's soldiers: $x = \frac{\sqrt{5}-1}{2}$, so $x^2 + x = 1$).
- Colouring:** chessboard, mod- k diagonal colouring $(i+j) \bmod k$, $(i \bmod 2, j \bmod 2)$ four-colouring.
- Monovariant** to prove termination: a positive integer that strictly decreases; lexicographic order; potential $\sum x_i^2$ (decreases under averaging moves).
- Semi-invariant** to prove unreachability: GCD of entries, determinant up to sign, number of inversions mod 2 (15-puzzle).



Left: removing opposite corners leaves 30 dark, 32 light squares; every domino covers one of each $\Rightarrow$ no tiling. Right: colouring by $(i+j) \bmod 4$; each 1×4 tile covers every colour once, so colour counts must be equal.

COUNTING

Core identities

Vandermonde $\sum_k \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$. Hockey stick $\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$. $\sum_k k \binom{n}{k} = n2^{n-1}$, $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$. Stars & bars: solutions of $x_1 + \dots + x_k = n$, $x_i \geq 0$: $\binom{n+k-1}{k-1}$.

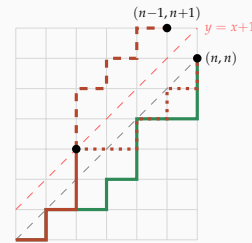
Inclusion–exclusion

$|\bigcup A_i| = \sum |A_i| - \sum |A_i \cap A_j| + \dots$. Derangements $D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!} = \text{round}(n!/e)$; surjections $[n] \rightarrow [k]$: $\sum_j (-1)^j \binom{k}{j} (k-j)^n = k! S(n, k)$. Bonferroni: truncating after a $+$ term overcounts.

Double counting. Count pairs (x, S) with $x \in S$ two ways: $\sum_S |S| = \sum_x d(x)$. Then apply C-S / Jensen to $\sum d(x)^2$ (counts pairs in a common set). Handshake: $\sum \deg = 2|E|$. Triangles: $\sum_v \binom{\deg v}{2}$ counts “cherries”.

Catalan numbers $C_n = \frac{1}{n+1} \binom{2n}{n}$

1, 1, 2, 5, 14, 42, 132, 429; $C_{n+1} = \sum_{i=0}^n C_i C_{n-i}$. Count: Dyck paths, balanced brackets, binary trees with n internal nodes, triangulations of an $(n+2)$ -gon, non-crossing partitions, 312-avoiding permutations.



Green: a good path (stays on/below $y = x$). Red: a bad path first touches $y = x+1$; reflecting the rest of it (dotted $\rightarrow$ dashed) gives a path to $(n-1, n+1)$, a bijection. So $C_n = \binom{2n}{n} - \binom{2n}{n+1}$.

Stirling, Bell, partitions

$S(n, k)$: partitions of $[n]$ into k blocks; $S(n, k) = kS(n-1, k) + S(n-1, k-1)$. Unsigned $c(n, k)$: permutations with k cycles; $\sum_k c(n, k) x^k = x(x+1) \cdots (x+n-1)$. Bell $B_{n+1} = \sum \binom{n}{k} B_k$. Integer partitions: $\sum p(n) x^n = \prod_k \frac{1}{1-x^k}$. **Euler:** partitions into distinct parts = into odd parts.

GENERATING FUNCTIONS

Toolkit

$\frac{1}{(1-x)^k} = \sum_n \binom{n+k-1}{k-1} x^n$. $\frac{1-\sqrt{1-4x}}{2x} = \sum C_n x^n$. $\frac{x}{1-x-x^2} = \sum F_n x^n$. Multiplying OGFs = convolving; $\frac{A(x)}{1-x} = \text{partial sums}$; $xA'(x) \leftrightarrow na_n$. EGF: e^x (sets), $\frac{1}{1-x}$ (orders), e^{e^x-1} (Bell), $\frac{e^{-x}}{1-x}$ (derangements).

Weighted/signed GFs for partitions of a set: “partition $\{1, \dots, n\}$ into two sets with equal sums of k -th powers” $\rightarrow \prod (1-x^i)$ and differentiate / evaluate at roots of unity. **Tiling / counting mod m** $\rightarrow$ roots of unity in $F(x)$.

GRAPH THEORY

Basic structure

Tree on n vertices: connected, $n - 1$ edges, unique paths. Cayley: n^{n-2} labelled trees. Euler circuit $\iff$ connected and all degrees even (trail: 0 or 2 odd vertices). Bipartite $\iff$ no odd cycle.

Planar graphs

$V - E + F = 2$ (connected). Simple planar, $V \geq 3$: $E \leq 3V - 6$; triangle-free: $E \leq 2V - 4$. So $K_5, K_{3,3}$ nonplanar; some vertex has degree ≤ 5 .

Matchings

Hall: bipartite (X, Y) has an X -saturating matching $\iff |N(S)| \geq |S|$ for all $S \subseteq X$. k -regular bipartite $\Rightarrow$ perfect matching (and edge-decomposes into k). **König**: in bipartite graphs, max matching = min vertex cover.

POSETS & SET SYSTEMS

Dilworth, Mirsky, Sperner, EKR

Min number of chains covering a poset = max antichain (Dilworth); dual: min antichains = longest chain (Mirsky). **Sperner**: an antichain in $2^{[n]}$ has size $\leq \binom{n}{\lfloor n/2 \rfloor}$; LYM: $\sum_A \binom{n}{|A|}^{-1} \leq 1$. **Erdős-Ko-Rado**: intersecting family of k -sets, $n \geq 2k$: size $\leq \binom{n-1}{k-1}$.

Linear-algebra method (over $\mathbb{F}_2$)

Oddtown: subsets of $[n]$ with odd sizes and pairwise even intersections: at most n (their indicator vectors are independent over $\mathbb{F}_2$). **Fisher**: if every two distinct sets meet in exactly $\lambda > 0$ elements, $m \leq n$. Principle: m vectors with Gram matrix of full rank in dimension $n \Rightarrow m \leq n$.

COMBINATORIAL GEOMETRY

- Pick**: lattice polygon: $A = I + \frac{B}{2} - 1$.
- Helly** ($\mathbb{R}^d$): convex sets, every $d + 1$ intersect $\Rightarrow$ all intersect.
- Sylvester-Gallai**: finitely many non-collinear points $\Rightarrow$ a line through exactly two.
- Convex hull / sweeping line**: order by x -coordinate or angle; rotate a line around a point ("windmill", IMO 2011/2).
- Any 5 points in general position contain a convex quadrilateral (Happy Ending).
- Lattice points: parity classes $(x \bmod 2, y \bmod 2) \Rightarrow$ among 5 points two have lattice midpoint.

MORE COMBINATORIAL TECHNIQUE

Greedy & algorithmic proofs. Describe a procedure and prove it works: it never gets stuck (an invariant) and it terminates (a monovariant). Greedy colouring gives $\chi \leq \Delta + 1$; greedy independent sets give $\alpha \geq \sum \frac{1}{d(v)+1}$ (Caro-Wei).

Induction by removal. Remove a vertex of min degree, a leaf, the largest element, the last row, a whole block. Check that the smaller object still satisfies the hypothesis (this is where most induction attempts fail). Sometimes remove *two* objects to keep parity.

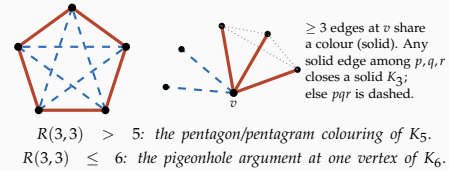
Model as a graph. People $\rightarrow$ vertices; "know each other" $\rightarrow$ edges. Cells $\rightarrow$ vertices; adjacency $\rightarrow$ edges. Tasks and workers $\rightarrow$ bipartite graph (Hall). Sequences of moves $\rightarrow$ paths in a state graph. Then ask about degrees, cycles, components, matchings, colourings.

Extremal graph theory

Mantel: triangle-free $\Rightarrow E \leq \lfloor n^2/4 \rfloor$. **Turán**: K_{r+1} -free $\Rightarrow E \leq (1 - \frac{1}{r}) \frac{n^2}{2}$, extremal = balanced complete r -partite. C_4 -free: $E \leq \frac{n}{4}(1 + \sqrt{4n-3})$.

Ramsey

$R(3,3) = 6$, $R(3,4) = 9$, $R(4,4) = 18$. $R(s,t) \leq R(s-1,t) + R(s,t-1)$, so $R(s,t) \leq \binom{s+t-2}{s-1}$. Lower bound $R(k,k) > 2^{k/2}$ (probabilistic).



Colouring & paths

Greedy: $\chi \leq \Delta + 1$; Brooks: $\chi \leq \Delta$ unless complete or odd cycle. Dirac: $\delta \geq n/2 \Rightarrow$ Hamiltonian cycle. Tournament always has a Hamiltonian path; a vertex reaching all others in ≤ 2 steps (king).

GAMES

Impartial games

Positions are P (previous player wins) or N. $P \iff$ every move goes to N. **Sprague-Grundy**: $g(x) = \text{mex}\{g(y)\}$; sum of games has $g = \oplus g_i$ (XOR). Nim: $P \iff$ XOR of heaps is 0.

Strategy stealing: if the second player had a winning strategy, the first could make an "extra" harmless move and adopt it (Chomp, Hex). **Pairing / symmetry**: mirror the opponent's move. **Invariant strategy**: maintain a property the opponent must break. Always check who moves last in terms of parity.

PROBABILISTIC METHOD

Expectation & LLL

$E[X] \geq c \Rightarrow$ some outcome has $X \geq c$. **Alteration**: pick randomly, then delete bad parts. **Lovász Local Lemma**: events A_i each with $P \leq p$, each independent of all but $\leq d$ others, $ep(d+1) \leq 1 \Rightarrow P(\bigcap A_i) > 0$.

Ex. every graph has a bipartite subgraph with $\geq E/2$ edges (random 2-colouring). Tournament with many Hamiltonian paths: $E = \frac{n!}{2^{n-1}}$.

PROCESS & ALGORITHM PROBLEMS

- Simulate small cases, look for a potential function.
- For "minimum number of moves": find a lower bound via an invariant that each move changes by at most 1.
- For "show the process terminates": monovariant in a well-ordered set.
- For "show some state is reached": periodicity + reversibility (a bijective process on a finite set is periodic, so returns to start).

Weights / potentials. Assign each object a weight so the target statement becomes "total weight > something". Weights 2^{-k} or x^k are standard; choose x so moves preserve or decrease the total.

Bijections. Before counting, ask whether the set is in bijection with something standard: lattice paths, binary strings, trees, Dyck words, compositions ($\binom{n-1}{k-1}$ compositions of n into k parts), permutations with a cycle structure.

Local modifications / exchange arguments. Take an optimal configuration; show that any "bad" local feature can be swapped out without making things worse. The optimum therefore has a clean structure.

Constructions. For "show N is achievable": start from extremal small cases, look for recursive (doubling, $n \rightarrow n+1$) or algebraic ($\mathbb{Z}/p$, finite fields, $\mathbb{F}_2^k$, cyclic shifts $i \mapsto i+k$) patterns. Symmetric constructions are easier to verify.

WORKED EXAMPLE · Pigeonhole on prefix sums
classical

Problem. Any n integers $a_1, \dots, a_n$ contain a nonempty block $a_i + \dots + a_j$ divisible by n .

Solution. Consider $S_0 = 0, S_1, \dots, S_n$ ($S_k = a_1 + \dots + a_k$): $n+1$ numbers, n residues mod n . Two agree, $S_i \equiv S_j$ with $i < j$, so $n \mid a_{i+1} + \dots + a_j$.

Lesson. *prefix sums turn “some block” into “two equal residues”.*

WORKED EXAMPLE · Colouring invariant
classical

Problem. Can a 10×10 board be tiled by 1×4 rectangles?

Solution. Colour cell (i, j) with $(i+j) \bmod 4$. Any 1×4 tile, horizontal or vertical, covers each colour exactly once, so a tiling needs equal colour counts. But the counts are 25, 26, 25, 24. No.

Lesson. *choose a colouring adapted to the tile’s shape so every tile covers a fixed pattern.*

WORKED EXAMPLE · Double counting
IMO 1998/2

Problem. a contestants, b judges ($b \geq 3$ odd); each judge rates each contestant pass/fail. Any two judges agree on at most k contestants. Show $\frac{k}{a} \geq \frac{b-1}{2b}$.

Solution. Count triples (contestant, unordered pair of judges agreeing on them). For a pair of judges: at most k , so $N \leq k \binom{b}{2}$. For a contestant passed by x judges: $\binom{x}{2} + \binom{b-x}{2}$ agreeing pairs, minimised at $x = \frac{b+1}{2}$ with value $\binom{b-1}{2}$. So $N \geq a \binom{b-1}{2}$. Combine: $\frac{k}{a} \geq \frac{\binom{b-1}{2}}{\binom{b}{2}} = \frac{2}{b(b-1)} = \frac{b-1}{2b}$.

Lesson. *count the object the hypothesis controls, from the side the conclusion is about.*

WORKED EXAMPLE · Cauchy–Schwarz on degrees
Mantel

Problem. A triangle-free graph on n vertices has at most $\frac{n^2}{4}$ edges.

Solution. For an edge uv , u and v have no common neighbour, so $d(u) + d(v) \leq n$. Summing over edges: $\sum_{uv \in E} (d(u) + d(v)) = \sum_v d(v)^2 \leq nE$. By C-S, $\sum d(v)^2 \geq \frac{(2E)^2}{n}$. So $\frac{4E^2}{n} \leq nE$, $E \leq \frac{n^2}{4}$.

Lesson. $\sum_{\text{edges}} (d(u) + d(v)) = \sum_v d(v)^2$ is the bridge between local and global.

WORKED EXAMPLE · Hall’s theorem
classical

Problem. A deck of 52 cards is dealt into 13 piles of 4. Show you can pick one card from each pile so that all 13 ranks appear.

Solution. Bipartite graph: piles vs. ranks, pile $\sim$ rank if the pile contains that rank. Any k piles hold $4k$ cards; each rank has only 4 cards, so these cards cover $\geq k$ ranks. Hall’s condition holds; a perfect matching picks the cards.

WORKED EXAMPLE · Roots-of-unity filter
classical

Problem. How many subsets of $\{1, \dots, 2000\}$ have sum divisible by 5?

Solution. $F(x) = \prod_{k=1}^{2000} (1 + x^k)$; answer $\frac{1}{5} \sum_{j=0}^4 F(\omega^j)$, $\omega = e^{2\pi i/5}$. Each residue class mod 5 occurs 400 times, so for $j \neq 0$: $F(\omega^j) = (\prod_{r=0}^4 (1 + \omega^r))^{400} = (2 \cdot 1)^{400}$, since $\prod_{r=1}^4 (1 + \omega^r) = \Phi_5(-1) = 1$. Answer $\frac{2^{2000} + 4 \cdot 2^{400}}{5}$.

Lesson. *“count with a condition mod m ” = generating function + roots of unity.*

WORKED EXAMPLE · Probabilistic method
Erdős 1947

Problem. Show $R(k, k) > 2^{k/2}$ for $k \geq 3$.

Solution. Colour K_n randomly, $n = \lfloor 2^{k/2} \rfloor$. Expected number of monochromatic K_k : $\binom{n}{k} 2^{1-\binom{k}{2}} < \frac{n^k}{k!} 2^{1-\frac{k(k-1)}{2}} \leq \frac{2^{k^2/2}}{k!} 2^{1-\frac{k^2}{2}+\frac{k}{2}} = \frac{2^{1+k/2}}{k!} < 1$. So some colouring has none.

Lesson. *if the expected number of bad events is < 1 , some outcome has none.*

WORKED EXAMPLE · Strategy stealing
Chomp

Problem. In an $m \times n$ chocolate bar ($mn > 1$) the bottom-left square is poisoned; players alternately choose a square and eat it with everything above and to the right; whoever eats the poison loses. Show the first player wins.

Solution. Suppose the second player wins. First player eats only the top-right square. The second player has a winning reply X . But the first player could have played X at the start (it also removes the top-right square), reaching the same position with the roles reversed. Contradiction. (The proof is non-constructive.)

COUNTING STRATEGIES AT A GLANCE

situation	strategy
“at least one”	complement, or PIE
arrangements with restrictions	fix the most constrained element first
circular arrangements	fix one element (divide by n) or Burnside
distributions	stars and bars; surjections via PIE/Stirling
paths in a grid	binomials, reflection, transfer matrix
recursive structure	recurrence, then GF or pattern
symmetric objects	Burnside’s lemma
“expected number”	linearity with indicators
parity of a count	pair up objects via an involution; count fixed points

Involutions. If σ is an involution on a finite set X , then $|X| \equiv |\text{Fix}(\sigma)| \pmod{2}$. Zagier’s one-sentence proof of the two-squares theorem is this idea. Sign-reversing involutions evaluate alternating sums: pair terms of opposite sign, count the unpaired ones.

V GEOMETRY

configurations · lemmas · transformations · coordinates

TRIANGLE TOOLKIT

Metric formulas (s semiperimeter, Δ area)

$$\frac{a}{\sin A} = 2R; a^2 = b^2 + c^2 - 2bc \cos A; \Delta = \frac{1}{2}ab \sin C = rs = \frac{abc}{4R} = \sqrt{s(s-a)(s-b)(s-c)}.$$

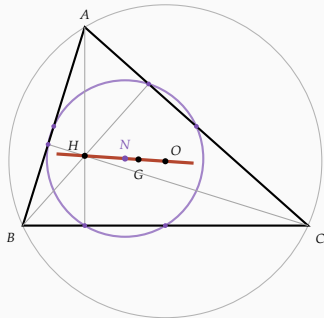
$$r_a = \frac{\Delta}{s-a}; \text{ tangent lengths from } A: \text{ incircle } s-a, \text{ A-excircle } s.$$

$$r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}; \cos A + \cos B + \cos C = 1 + \frac{r}{R}.$$

$$\text{Median } m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4}. \text{ Bisector divides } BC \text{ in ratio } c : b.$$

$$\text{Stewart (D on BC, BD = m, DC = n, AD = d): } b^2m + c^2n = a(d^2 + mn).$$

$$\text{Euler: } OI^2 = R^2 - 2Rr \text{ (so } R \geq 2r); OI_A^2 = R^2 + 2Rr_a. OH^2 = 9R^2 - (a^2 + b^2 + c^2). AH = 2R \cos A.$$



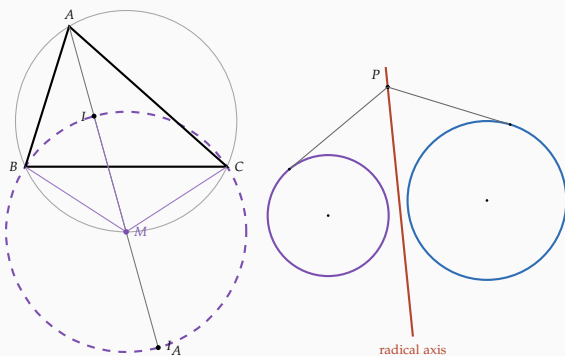
Euler line O, G, H with $OG : GH = 1 : 2$; nine-point circle (centre $N = \text{midpoint of } OH$, radius $R/2$) passes through the 3 feet of altitudes, 3 midpoints, and midpoints of AH, BH, CH .

Orthocentre facts

Reflections of H over the sides and over the side midpoints lie on (ABC) ; the latter are the antipodes A' of A , and $HBA'C$ is a parallelogram. O and H are isogonal conjugates; $\angle BAH = \angle OAC = 90^\circ - B$.

Incenter–excenter lemma

Let AI meet (ABC) again at M (midpoint of arc BC). Then $MB = MC = MI = MI_A$; B, I, C, I_A lie on a circle centred at M .



Left: incenter–excenter lemma. Right: the radical axis is the locus of equal power; tangents from P are equal.

CIRCLES

Power of a point

$\text{Pow}(P) = PO^2 - r^2 = PA \cdot PB$ for any line through P meeting the circle at A, B (signed) $= PT^2$ for a tangent.

Radical axis of two circles: $\{\text{Pow}_1 = \text{Pow}_2\}$, a line $\perp$ to O_1O_2 (the common chord if they meet). Three circles: axes concur at the **radical centre** (or are parallel).

Cyclic quadrilaterals

$ABCD$ cyclic $\iff \angle ABD = \angle ACD \iff \angle A + \angle C = 180^\circ \iff PA \cdot PC = PB \cdot PD$ ($P = AC \cap BD$) $\iff$ **Ptolemy**: $AC \cdot BD = AB \cdot CD + AD \cdot BC$ (in general $\leq$; equality iff cyclic). Tangent–chord: angle between tangent and chord = inscribed angle on the other side.

Use **directed angles** $\angle(\ell, m) \bmod 180^\circ$ to make angle chases configuration-independent: A, B, C, D concyclic $\iff \angle ABC = \angle ADC$.

Miquel & spiral similarity

Points on the sides of ABC : the three circles $(AEF), (BFD), (CDE)$ concur (Miquel point). For a complete quadrilateral, the four triangle circumcircles concur at the Miquel point M . If $AB \rightarrow CD$ by a spiral similarity, its centre is the second intersection of (XAB) and (XCD) , where $X = AC \cap BD$; the same centre also sends $AC \rightarrow BD$ (spiral similarities come in pairs).

Simson line

The feet of perpendiculars from P to the sides of ABC are collinear $\iff P \in (ABC)$.

CONCURRENCE & COLLINEARITY

Ceva, Menelaus

Cevians AD, BE, CF concur $\iff \frac{BD}{DC} \frac{CE}{EA} \frac{AF}{FB} = 1$ (signed) $\iff \frac{\sin \angle BAD}{\sin \angle DAC} \frac{\sin \angle CBE}{\sin \angle EBA} \frac{\sin \angle ACF}{\sin \angle FCB} = 1$ (trig Ceva). Menelaus: D, E, F collinear on the side lines $\iff$ the signed product $= -1$.

$$\text{Ratio lemma: } \frac{BD}{DC} = \frac{AB \sin \angle BAD}{AC \sin \angle DAC}.$$

Projective classics

Pascal: for a hexagon on a conic, intersections of opposite sides are collinear (degenerate versions with tangents: $AABBCC$). **Brianchon** (dual). **Desargues**: triangles perspective from a point $\iff$ from a line. **Pappus**.

Harmonic bundles

$(A, B; C, D) = \frac{CA}{CB} / \frac{DA}{DB}$; harmonic if $= -1$. Preserved by perspectivities (projecting from a point onto another line, or onto a circle through that point). Standard sources: complete quadrilaterals; $(B, C; D, D') = -1$ for the feet of the internal/external bisectors from A ; if PC, PD are tangents to a circle and a line through P meets it at A, B , then $ACBD$ is harmonic. If $(A, B; C, D) = -1$ and M is the midpoint of AB : $MC \cdot MD = MA^2$.

Symmedian. Reflection of the median over the angle bisector. The A -symmedian passes through the intersection of tangents to (ABC) at B, C ; the symmedian point is $K = (a^2 : b^2 : c^2)$. $ABDC$ with AD a symmedian chord is a *harmonic quadrilateral*: $AB \cdot CD = AC \cdot BD$.

TRANSFORMATIONS

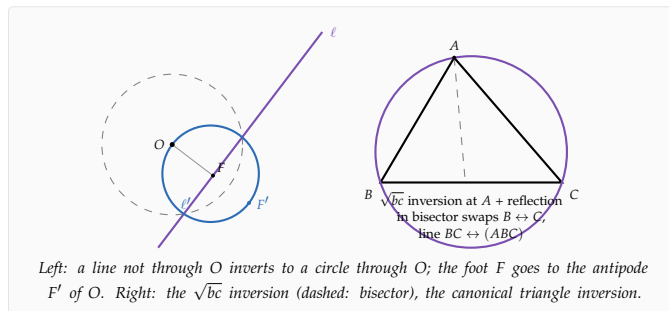
Homothety

Two circles have an external and internal centre of homothety. **Monge:** the three external centres of three circles are collinear. Tangent circles: the point of tangency is a homothety centre, so tangency point, the two centres, and corresponding points align.

Incircle diameter lemma. If the incircle touches BC at D , DD' is a diameter, and AD' meets BC at E , then E is the A -excircle touch point, $BD = CE = s - b$, so the midpoint of BC is the midpoint of DE (homothety at A sends the incircle to the A -excircle).

Inversion (centre O , radius r)

$P \mapsto P'$ on ray OP with $OP \cdot OP' = r^2$. Lines through $O \leftrightarrow$ themselves; lines not through $O \leftrightarrow$ circles through O ; circles not through $O \leftrightarrow$ circles. Angles preserved (orientation reversed). $P'Q' = \frac{r^2 PQ}{OP \cdot OQ}$. A circle is fixed iff orthogonal to the circle of inversion (or it is that circle).



► **SIGNAL:** tangent circles, many circles through one point, “prove a circle is tangent to...”, products of lengths. Invert at the point where the most circles meet.

HOW TO ATTACK A GEOMETRY PROBLEM

1. **Draw two accurate diagrams** (different shapes). Mark every equal angle you see; conjecture collinearities and concyclicities.
2. **Identify the configuration.** Incircle? Orthocentre? Arc midpoints? Tangent circles? Match to the lemma list.
3. **Angle chase** with directed angles. Most “prove concyclic/tangent” problems start here.
4. **Find the right auxiliary point:** arc midpoint, antipode, reflection of H , Miquel point, the second intersection of two natural circles.
5. **Phantom points:** to show X lies on ℓ , define X' as the intersection that should work, prove it has X ’s defining property, conclude $X = X'$.
6. **Transform:** homothety for tangent circles, inversion for many circles through a point, spiral similarity for two “similar” triangles sharing a vertex, reflection for angle bisectors.
7. **Compute** (complex, barycentric, trig) if synthetic stalls after 40 min, choosing the system that makes the given data simplest.

Angle-chasing checklist

- Inscribed angles on the same chord; tangent–chord; cyclic quadrilateral opposite angles.
- Isosceles triangles from equal radii ($OA = OB$), tangent lengths, arc midpoints ($MB = MC$).
- $\angle BIC = 90^\circ + \frac{A}{2}$, $\angle BHC = 180^\circ - A$, $\angle BOC = 2A$, $\angle BIA = 90^\circ - \frac{A}{2}$.
- Right angles: feet of altitudes lie on circles with diameters AB, AH , etc.
- Parallel lines transfer angles; spiral similarities preserve them.

COORDINATES, WHEN SYNTHETIC FAILS

Complex numbers (unit circle $|a| = 1$, $\bar{a} = 1/a$)

Chord ab : $z + ab\bar{z} = a + b$. Foot from z to chord ab : $\frac{1}{2}(a + b + z - ab\bar{z})$. Tangents at a, b meet at $\frac{2ab}{a+b}$. For a, b, c on the unit circle: $H = a + b + c$, $G = \frac{a+b+c}{3}$, $O = 0$. Incentre: set $a = x^2, b = y^2, c = z^2$ with suitable signs; $I = -(xy + yz + zx)$. a, b, c collinear $\iff \frac{a-c}{b-c} \in \mathbb{R}$; a, b, c, d concyclic $\iff$ cross-ratio $\frac{(a-c)(b-d)}{(b-c)(a-d)} \in \mathbb{R}$. $AB \perp CD \iff \frac{a-b}{c-d} \in i\mathbb{R}$.

Barycentric coordinates ($x : y : z$), $x + y + z = 1$

$G = (1 : 1 : 1)$, $I = (a : b : c)$, $I_A = (-a : b : c)$, $K = (a^2 : b^2 : c^2)$, $O = (a^2 S_A : \dots) = (\sin 2A : \sin 2B : \sin 2C)$, $H = (S_B S_C : \dots) = (\tan A : \tan B : \tan C)$, Conway $S_A = \frac{b^2 + c^2 - a^2}{2}$. Circumcircle: $a^2 yz + b^2 zx + c^2 xy = 0$. Line: $ux + vy + wz = 0$. Collinearity/area: $[PQR]/[ABC] = \det$ of the normalized rows. Distance for displacement (x, y, z) with $x + y + z = 0$: $|PQ|^2 = -a^2 yz - b^2 zx - c^2 xy$. Perpendicularity (displacements M, N): $a^2(z_1 y_2 + y_1 z_2) + b^2(x_1 z_2 + z_1 x_2) + c^2(y_1 x_2 + x_1 y_2) = 0$.

Which bash? Circle-heavy with a given circumcircle $\rightarrow$ complex. Cevians, ratios, triangle centres $\rightarrow$ barycentric. Two circles and lines $\rightarrow$ Cartesian with smart axes. Angles galore $\rightarrow$ trig (Ceva/ratio lemma). Budget the time; half a bash scores 0.

CONFIGURATION LEMMAS TO RECOGNISE

- **Reim:** two circles meet at P, Q ; lines through P and Q cut them at A, C and B, D : $AB \parallel CD$.
- **Butterfly:** M midpoint of chord PQ ; chords AB, CD through M ; AD, BC cut PQ at X, Y : $MX = MY$.
- **Brocard:** for cyclic $ABCD$, O is the orthocentre of the diagonal triangle.
- **Mixtilinear:** the A -mixtilinear incircle touches AB, AC at points whose midpoint is I .
- **Feuerbach:** nine-point circle is tangent to incircle and all excircles.
- **Isogonal conjugates:** P, P^* have the same pedal circle; $O \leftrightarrow H$, $G \leftrightarrow K$.
- **Sawayama–Thébault, Poncelet porism, Newton line, Gauss line:** recognise, don’t reprove.

More lemmas

Shooting lemma

M the midpoint of arc BC of (ABC) not containing A . A line through M meets BC at X and the circle again at Y . Then $MX \cdot MY = MB^2$ (so (BXY) is tangent to MB).

Right angles on the incircle chord

Incicle touches CA, AB at E, F ; M, N midpoints of BC, CA . Then BI, EF and MN concur at a point K with $\angle BKC = 90^\circ$.

Lemmas on the A -excircle and incircle

Incicle touches BC at D ($BD = s - b$); A -excircle at D' ($BD' = s - c$); so D, D' are symmetric about the midpoint M of BC . If X is the antipode of D on the incircle, A, X, D' are collinear (diameter lemma), and $MI \parallel AD'$ (midline of triangle DXD').

Reflections & isogonals

The reflections of a line through H over the sides concur on (ABC) (the anti-Steiner point). Isogonal lines at A : AO and AH ; the median and the symmedian.

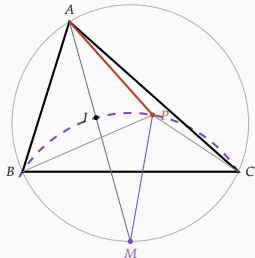
WORKED EXAMPLE · Incenter–excenter lemma

IMO 2006/1

Problem. I is the incentre of ABC ; P inside with $\angle PBA + \angle PCA = \angle PBC + \angle PCB$. Show $AP \geq AI$, with equality iff $P = I$.

Solution. The condition says $\angle PBC + \angle PCB = \frac{B+C}{2}$, so $\angle BPC = 180^\circ - \frac{B+C}{2} = 90^\circ + \frac{A}{2} = \angle BIC$. Hence P lies on circle (BIC) , whose centre is the arc midpoint M (incenter–excenter lemma), and A, I, M are collinear with I between. By the triangle inequality $AP \geq AM - MP = AM - MI = AI$, equality iff P is on segment AM , i.e. $P = I$.

Lesson. an angle condition often places a point on a known circle; the lemma supplies the centre.



P moves on the arc of (BIC) centred at M ; AP is shortest at I .

WORKED EXAMPLE · Radical axis

IMO 2008/1

Problem. H is the orthocentre of acute ABC . The circle centred at the midpoint of BC through H meets line BC at A_1, A_2 ; define B_1, B_2, C_1, C_2 similarly. Show these six points are concyclic.

Solution. Let M_b, M_c be midpoints of CA, AB . The circles centred M_b, M_c through H both pass through H ; their radical axis passes through H and is $\perp M_b M_c \parallel BC$, so it is the altitude AH . Hence A has equal power: $AB_1 \cdot AB_2 = AC_1 \cdot AC_2$, so B_1, B_2, C_1, C_2 are concyclic. Their circle's centre lies on the perpendicular bisectors of $B_1 B_2$ (the perpendicular to CA at M_b) and $C_1 C_2$: it is the circumcentre O . Likewise C_1, C_2, A_1, A_2 and A_1, A_2, B_1, B_2 lie on circles centred at O . Circles with a common centre that share a point (Ex. C_1) coincide, so all six points lie on one circle.

Lesson. “equal power from a vertex” is the standard route to four concyclic points on two lines.

WORKED EXAMPLE · Complex numbers

classical

Problem. Show that the reflection of H over BC lies on (ABC) .

Solution. Unit circle, $h = a + b + c$. Reflection of z over chord bc is $b + c - bcz$. With $\bar{h} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$: $b + c - bc(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}) = -\frac{bc}{a}$, which has modulus 1.

WORKED EXAMPLE · Inversion

Ptolemy's inequality

Problem. For any four points, $AB \cdot CD + AD \cdot BC \geq AC \cdot BD$, with equality iff $ABCD$ is cyclic (in order).

Solution. Invert at A with radius 1: $B'C' = \frac{BC}{AB \cdot AC}$ etc. The triangle inequality $B'C' + C'D' \geq B'D'$ becomes $\frac{BC}{AB \cdot AC} + \frac{CD}{AC \cdot AD} \geq \frac{BD}{AB \cdot AD}$; multiply by $AB \cdot AC \cdot AD$. Equality iff B', C', D' are collinear in that order, i.e. B, C, D lie on a circle through A in order.

Lesson. inversion turns products/quotients of distances into ordinary distances.

WORKED EXAMPLE · Trig Ceva

classical

Problem. Show the altitudes are concurrent.

Solution. For the altitude from A : $\angle BAD = 90^\circ - B$, $\angle DAC = 90^\circ - C$. The trig Ceva product is $\frac{\cos B}{\cos C} \cdot \frac{\cos C}{\cos A} \cdot \frac{\cos A}{\cos B} = 1$.

Lesson. trig Ceva makes concurrency of angle-defined cevians a one-liner.

VI THE PUTNAM TOOLKIT

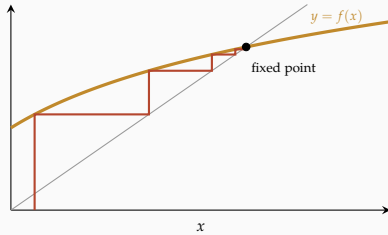
analysis · linear algebra · abstract algebra · probability

LIMITS & SEQUENCES

Core theorems

Monotone + bounded $\Rightarrow$ convergent. Bolzano–Weierstrass. Cauchy criterion. Squeeze. **Stolz–Cesàro**: $b_n \uparrow \infty$, $\frac{a_{n+1}-a_n}{b_{n+1}-b_n} \rightarrow L \Rightarrow \frac{a_n}{b_n} \rightarrow L$. Hence Cesàro means: $a_n \rightarrow L \Rightarrow \frac{1}{n} \sum a_k \rightarrow L$. Ratio $\Rightarrow$ root: $\frac{a_{n+1}}{a_n} \rightarrow L \Rightarrow a_n^{1/n} \rightarrow L$. Fekete: subadditive $a_{m+n} \leq a_m + a_n \Rightarrow \frac{a_n}{n} \rightarrow \inf \frac{a_n}{n}$.

Iterated maps $x_{n+1} = f(x_n)$. Find fixed points, show an interval is invariant, use monotonicity (f increasing $\Rightarrow x_n$ monotone) or contraction ($|f'| \leq q < 1$). **Rate**: if $x_{n+1} = x_n - ax_n^{k+1} + \dots \rightarrow 0$, then x_n^{-k} grows like kan (apply Stolz to x_n^{-k}). Ex. $x_{n+1} = \sin x_n$: $x_n \sim \sqrt{3/n}$.



Cobweb: f increasing with $0 < f' < 1$ near the fixed point $\Rightarrow$ monotone convergence.

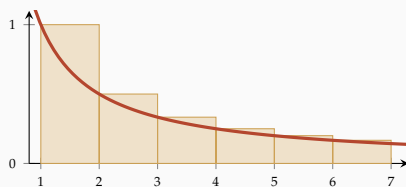
Asymptotics to know

$H_n = \ln n + \gamma + \frac{1}{2n} + O(n^{-2})$; $n! \sim \sqrt{2\pi n} (n/e)^n$; $\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}$; $(1 + \frac{x}{n})^n \rightarrow e^x$; $\sum_{k \leq n} k^p \sim \frac{n^{p+1}}{p+1}$; Wallis $\prod \frac{4k^2}{4k^2-1} = \frac{\pi}{2}$; $\pi(n) \sim \frac{n}{\ln n}$.

SERIES

Convergence tests

Comparison / limit comparison; ratio; root; integral test ($f \searrow 0$: $\sum f(n)$ and $\int f$ converge together); **Cauchy condensation** ($a_n \downarrow$: $\sum a_n \sim \sum 2^k a_{2^k}$); alternating (Leibniz); **Dirichlet**: $a_n \downarrow 0$ and partial sums of b_n bounded $\Rightarrow \sum a_n b_n$ converges (so $\sum \frac{\sin n}{n}$ converges). $\sum \frac{1}{n^p}$, $\sum \frac{1}{n(\ln n)^p}$ converge iff $p > 1$.



Upper sums vs. integral: $\ln(n+1) < H_n < 1 + \ln n$. The same picture proves every integral-test bound.

Sums worth memorising

$\sum \frac{1}{n^2} = \frac{\pi^2}{6}$, $\sum \frac{1}{n^4} = \frac{\pi^4}{90}$, $\sum \frac{(-1)^{n+1}}{n} = \ln 2$, $\sum \frac{(-1)^n}{2n+1} = \frac{\pi}{4}$, $\sum \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$, $\sum_{n \geq 1} \frac{1}{n^2+n} = 1$, $\sum \frac{n}{2^n} = 2$, $\sum_{n \geq 0} \frac{x^n}{n!} = e^x$, $\sum nx^n = \frac{x}{(1-x)^2}$, $\sum \binom{2n}{n} x^n = \frac{1}{\sqrt{1-4x}}$.

Power-series trick. Recognise $\sum a_n$ as $F(1)$ or $F(\frac{1}{2})$ for a known F ; differentiate/integrate termwise inside the radius. Abel's theorem lets you take $x \rightarrow 1^-$ if the series converges at 1. Swap sum/integral: $\frac{1}{n+1} = \int_0^1 x^n dx$; $\frac{1}{n^2} = -\int_0^1 x^{n-1} \ln x dx$.

CALCULUS THEOREMS

Mean values & friends

IVT; Rolle; MVT; **Cauchy MVT** $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(\xi)}{g'(\xi)}$; **Darboux**: derivatives have the intermediate-value property (even if discontinuous); Taylor with Lagrange remainder $\frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1}$; L'Hôpital.

Auxiliary-function trick. To show $f'(\xi) = \lambda f(\xi)$ somewhere, apply Rolle to $e^{-\lambda x} f(x)$. For $f'(\xi) + g'(\xi)f(\xi) = 0$: use $e^{\int g}$. For " $f(a) = f(b) = 0 \Rightarrow \exists \xi : f'(\xi) = f(\xi)^2$ " etc., find the integrating factor.

Continuous $\neq$ differentiable; pointwise limits of continuous functions need not be continuous (uniform convergence needed to swap lim and $\int$).

INTEGRALS

Tricks

King's rule: $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$. Average the two versions: $\int_0^{\pi/2} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{4}$. $x \mapsto \frac{1}{x}$ on $(0, \infty)$: $\int_0^\infty \frac{\ln x}{1+x^2} dx = 0$.

Feynman: differentiate $I(t)$ under the integral: $\int_0^1 \frac{x^t-1}{\ln x} dx = \ln(t+1)$. **Frullani**: $\int_0^\infty \frac{f(ax)-f(bx)}{x} dx = (f(0)-f(\infty)) \ln \frac{b}{a}$. **Riemann sums**: $\frac{1}{n} \sum_{k=1}^n f(\frac{k}{n}) \rightarrow \int_0^1 f$; turn $\lim \sqrt[n]{\dots}$ into $\exp \int \ln$.

Standard values

$\int_{-\infty}^\infty e^{-x^2} = \sqrt{\pi}$; $\int_0^\infty \frac{\sin x}{x} = \frac{\pi}{2}$; $\int_0^1 x^m (1-x)^n dx = \frac{m!n!}{(m+n+1)!}$; $\Gamma(\frac{1}{2}) = \sqrt{\pi}$; $\int_0^{\pi/2} \ln \sin x dx = -\frac{\pi}{2} \ln 2$; $\int_0^{\pi/2} \sin^{2n} x dx = \frac{\pi}{2} \frac{(2n-1)!!}{(2n)!!}$; $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx = \frac{\pi}{8} \ln 2$; $\int_0^\infty \frac{x^{s-1}}{e^x-1} dx = \Gamma(s)\zeta(s)$.

Integral inequalities

C-S $(\int fg)^2 \leq \int f^2 \int g^2$; Jensen $\varphi(\frac{1}{b-a} \int f) \leq \frac{1}{b-a} \int \varphi \circ f$ (φ convex); Chebyshev: f, g both increasing $\Rightarrow \int_0^1 fg \geq \int_0^1 f \int_0^1 g$; Hermite–Hadamard $f(\frac{a+b}{2}) \leq \frac{1}{b-a} \int_a^b f \leq \frac{f(a)+f(b)}{2}$ (f convex).

LINEAR ALGEBRA

Determinants

Multiplicative; $\det A^T = \det A$; alternating multilinear in rows. Vandermonde $\det[x_i^{j-1}] = \prod_{i < j} (x_j - x_i)$. Circulant with first row $(c_0, \dots, c_{n-1})$: eigenvalues $\sum c_k \omega^{jk}$. **Matrix determinant lemma**: $\det(A + uv^T) = \det A (1 + v^T A^{-1}u)$. Block: $\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \det(D - CA^{-1}B)$. **Hadamard**: $|\det A| \leq \prod \|\text{rows}\|$. $\det(J - I) = (-1)^{n-1}(n-1)$ for the all-ones J .

Parity trick. To show $\det \neq 0$ for a $0/\pm 1$ matrix, compute it mod 2 (permanent $\equiv \det \bmod 2$). Ex. an $n \times n$ matrix, n even, with 0 diagonal and ± 1 elsewhere is $\equiv J - I \pmod{2}$, and $\det(J - I) = (-1)^{n-1}(n-1)$ is odd, so it is invertible.

Rank, trace, eigenvalues

Rank-nullity; $\text{rank}(AB) \leq \min$; Sylvester $\text{rank } A + \text{rank } B - n \leq \text{rank}(AB)$; $\text{rank}(A+B) \leq \text{rank } A + \text{rank } B$. $\text{tr}(AB) = \text{tr}(BA)$; $\text{tr } A = \sum \lambda_i$, $\det A = \prod \lambda_i$; AB and BA have the same characteristic polynomial. **Cayley–Hamilton**: $p_A(A) = 0$. Idempotent: $\text{rank} = \text{tr}$. Nilpotent $\Leftrightarrow$ all eigenvalues 0 $\Leftrightarrow \text{tr } A^k = 0 \forall k$ (char. o). Real symmetric: orthogonally diagonalisable, real eigenvalues. Gram matrices are PSD; rank of m vectors' Gram matrix $\leq$ dimension.

Counting via rank. Encode a combinatorial object as a matrix; rank $\leq n$ bounds the number of objects (see Oddtown). Eigenvalues of the adjacency matrix: $\lambda_{\max} \geq$ average degree; #closed walks of length $k = \text{tr } A^k$.

ABSTRACT ALGEBRA

Groups

Lagrange: $|H|$ divides $|G|$; so $\text{ord}(g) \mid |G|$ and $g^{|G|} = e$. **Cauchy:** $p \mid |G| \Rightarrow$ element of order p . $x^2 = e \ \forall x \Rightarrow G$ abelian. $|G| = p^2 \Rightarrow$ abelian. Every group of prime order is cyclic. Class equation: $|G| = |Z(G)| + \sum [G : C(x)] \Rightarrow p$ -groups have nontrivial centre. Orbit-stabiliser $|G| = |\text{Orb}(x)| |\text{Stab}(x)|$; **Burnside:** $\# \text{orbits} = \frac{1}{|G|} \sum_g |\text{Fix}(g)|$ (necklace counting).

Fields & polynomials mod p

$\mathbb{F}_p^\times$ cyclic. $(a+b)^p = a^p + b^p$ in characteristic p . $\mathbb{F}_{p^n}$ exists and is unique; $x^{p^n} - x = \prod$ monic irreducibles over $\mathbb{F}_p$ of degree dividing n . A degree-2 or 3 polynomial is irreducible iff it has no root. $\mathbb{Z}[x]$: irreducible mod p (with same degree) $\Rightarrow$ irreducible over $\mathbb{Q}$.

PROBABILITY

Principles

Linearity of expectation (no independence needed) with indicator variables. $E[X] = \sum_{k \geq 1} P(X \geq k)$ for $X \in \mathbb{N}$. Geometric waiting time $E = \frac{1}{p}$. Coupon collector nH_n . Conditioning / first-step analysis: set up linear equations for E_s over states s .

Classics

Fair gambler's ruin from i to $\{0, N\}$: $P(\text{hit } N) = \frac{i}{N}$, expected time $i(N-i)$. n uniform points on a circle lie in a semicircle w.p. $\frac{n}{2^{n-1}}$. Broken stick (2 cuts) forms a triangle w.p. $\frac{1}{4}$. k -th smallest of n uniforms has mean $\frac{k}{n+1}$. Random permutation: expected fixed points 1, cycles H_n ; $P(\text{derangement}) \rightarrow \frac{1}{e}$. Buffon's needle ($\ell \leq d$): $\frac{2\ell}{\pi d}$.

MORE PUTNAM TECHNIQUE

Compactness & extreme values. A continuous function on a closed bounded set attains its max and min; use it to *define* an extremal point, then derive properties there (derivative zero, or boundary). This is the analyst's extremal principle.

Fixed-point thinking. A continuous $f : [a, b] \rightarrow [a, b]$ has a fixed point (IVT on $f(x) - x$). Contractions have unique fixed points and iterates converge.

Symmetrise integrals. $x \mapsto a+b-x$, $x \mapsto \frac{1}{x}$, $x \mapsto \frac{\pi}{2} - x$, $x \mapsto -x$ (odd/even parts), $\theta \mapsto \frac{\pi}{4} - \theta$ after $x = \tan \theta$. Add the original and transformed integrals.

Turn discrete into continuous. Sums $\rightarrow$ integrals (Riemann sums, Euler-Maclaurin), recurrences $\rightarrow$ differential equations (heuristically, then make rigorous), products $\rightarrow$ exponentials of sums of logs.

Linear algebra in disguise. "Show the numbers can be chosen" $\rightarrow$ solve a linear system (nonzero determinant). "At most n such sets" $\rightarrow$ linear independence. "Recurrence" $\rightarrow$ powers of a companion matrix. "Quadratic form positive" $\rightarrow$ eigenvalues / Gram matrix.

Polynomials with calculus. Rolle: between two roots of P lies a root of P' ; if P has n real roots (with multiplicity), P' has at least $n-1$. Gauss-Lucas: roots of P' lie in the convex hull of roots of P . $\frac{P'}{P} = \sum \frac{1}{x-r_i}$ turns root questions into sums.

Pigeonhole for reals. Among $n+1$ numbers in $[0, 1)$, two are within $\frac{1}{n}$. Dirichlet approximation: $|q\alpha - p| < \frac{1}{n}$ for some $1 \leq q \leq n$.

Swapping limits, sums and integrals requires justification (monotone/dominated convergence, uniform convergence, absolute convergence for rearranging series). The Putnam graders do check.

Symmetry first. Before computing, ask what is exchangeable: Ex. the probability that the k -th card drawn is an ace is $\frac{4}{52}$ regardless of k . **Geometric probability:** draw the sample space as a region (square, simplex) and compute areas.

PUTNAM PATTERNS

- A1/B1:** often a clever observation, small cases, parity, or a direct computation. Do these first and write them perfectly.
- "Find all n ":** compute through $n \approx 10$ by hand; the answer is usually a simple family or finite list.
- Polynomial problems:** roots, Vieta, degree comparison, evaluate at roots of unity, derivative (P' has roots between roots of P , Gauss-Lucas: in the convex hull).
- Functional/limit hybrids:** continuity + density of $\mathbb{Q}$ or of $\{m + n\alpha\}$.
- Matrices over $\mathbb{F}_2$ or $\mathbb{Z}$:** work mod p , use determinants/rank.
- Games on integers:** compute P-positions for small cases; look for a pattern in binary.
- Integer-valued/limit:** $\lfloor \cdot \rfloor$ and fractional parts; equidistribution.
- Geometry:** vectors, coordinates, and calculus are allowed; use them.

Putnam answers with a numerical value must be *justified*; an unproved correct answer usually scores 0-1.

FURTHER READING

- Engel, *Problem-Solving Strategies*; Zeitz, *The Art and Craft of Problem Solving*.
- Evan Chen, *Euclidean Geometry in Mathematical Olympiads* (EGMO); *The OTIS Excerpts*.
- Pham Kim Hung, *Secrets in Inequalities*; Lee, *Topics in Inequalities*.
- Andreescu & Dospinescu, *Problems from the Book; Straight from the Book*.
- Gelca & Andreescu, *Putnam and Beyond*; Kedlaya-Poonen-Vakil, *The Putnam Competition 1985-2000*.
- IMO Shortlists (the best problem bank there is), AoPS forum solutions.

WORKED EXAMPLE · King's rule

Putnam 1980 A3

Problem. Evaluate $I = \int_0^{\pi/2} \frac{dx}{1 + (\tan x)^{\sqrt{2}}}$.

Solution. Under $x \mapsto \frac{\pi}{2} - x$, $\tan x \mapsto \cot x$ and the integrand becomes $\frac{(\tan x)^{\sqrt{2}}}{1 + (\tan x)^{\sqrt{2}}}$. Adding: $2I = \int_0^{\pi/2} 1 dx$, so $I = \frac{\pi}{4}$.

Lesson. a weird exponent is a decoy; the symmetry does not care about it.

WORKED EXAMPLE · Substitution + King

Putnam 2005 A5

Problem. Evaluate $\int_0^1 \frac{\ln(x+1)}{x^2+1} dx$.

Solution. $x = \tan \theta$: $I = \int_0^{\pi/4} \ln(1 + \tan \theta) d\theta$. Now $\theta \mapsto \frac{\pi}{4} - \theta$: $1 + \tan(\frac{\pi}{4} - \theta) = \frac{2}{1 + \tan \theta}$. So $I = \int_0^{\pi/4} (\ln 2 - \ln(1 + \tan \theta)) d\theta = \frac{\pi}{4} \ln 2 - I$, giving $I = \frac{\pi}{8} \ln 2$.

WORKED EXAMPLE · Rate of convergence

classical

Problem. $x_1 \in (0, \pi)$, $x_{n+1} = \sin x_n$. Find $\lim \sqrt{n} x_n$.

Solution. $x_n \downarrow 0$ (as $0 < \sin x < x$). $\frac{1}{\sin^2 x} - \frac{1}{x^2} = \frac{x^2 - \sin^2 x}{x^2 \sin^2 x} \rightarrow \frac{1}{3}$ as $x \rightarrow 0$ (expand $\sin^2 x = x^2 - \frac{x^4}{3} + \dots$). So $\frac{1}{x_{n+1}^2} - \frac{1}{x_n^2} \rightarrow \frac{1}{3}$ and by Stolz-Cesàro $\frac{1}{nx_n^2} \rightarrow \frac{1}{3}$. Answer $\sqrt{3}$.

Lesson. find a function g with $g(x_{n+1}) - g(x_n) \rightarrow \text{const}$; here $g = x^{-2}$.

WORKED EXAMPLE · Riemann sums in a limit

classical

Problem. Find $\lim \frac{\sqrt[n]{n!}}{n}$.

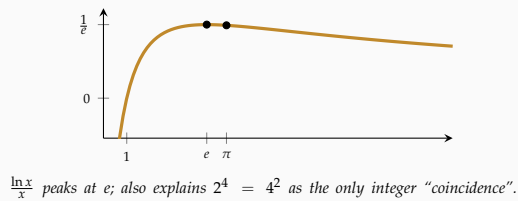
Solution. $\ln \frac{\sqrt[n]{n!}}{n} = \frac{1}{n} \sum_{k=1}^n \ln \frac{k}{n} \rightarrow \int_0^1 \ln x dx = -1$ (improper but monotone, so fine). Limit $\frac{1}{e}$.

WORKED EXAMPLE · Comparing e^π and π^e

classical

Problem. Which is larger, e^π or π^e ?

Solution. Compare $\frac{\ln e}{e}$ and $\frac{\ln \pi}{\pi}$. $g(x) = \frac{\ln x}{x}$ has $g' = \frac{1 - \ln x}{x^2}$, maximum at $x = e$. So $\frac{\ln \pi}{\pi} < \frac{1}{e}$, i.e. $e \ln \pi < \pi$: $\pi^e < e^\pi$.



WORKED EXAMPLE · Complex factorisation of matrices classical

Problem. Real $n \times n$ matrices satisfy $A^2 + B^2 = AB$ and $BA - AB$ is invertible. Show $3 \mid n$.

Solution. Let $\omega = e^{2\pi i/3}$. Then $(A + \omega B)(A + \bar{\omega}B) = A^2 + B^2 + \bar{\omega}AB + \omega BA = (1 + \bar{\omega})AB + \omega BA = \omega(BA - AB)$, using $1 + \bar{\omega} = -\omega$. The left determinant is $\det(A + \omega B)\overline{\det(A + \omega B)} \geq 0$, real. So $\omega^n \det(BA - AB)$ is real, and $\det(BA - AB)$ is a nonzero real: $\omega^n \in \mathbb{R}$, so $3 \mid n$.

Lesson. factor over $\mathbb{C}$ with conjugate pairs to create $|\det|^2$.

WORKED EXAMPLE · Power series classical

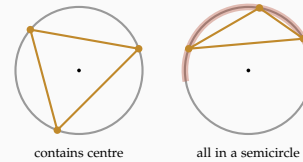
Problem. Evaluate $\sum_{n \geq 1} \frac{1}{n2^n}$.

Solution. $-\ln(1-x) = \sum \frac{x^n}{n}$ for $|x| < 1$; at $x = \frac{1}{2}$: $\ln 2$.

WORKED EXAMPLE · Geometric probability Putnam 1992 A6 (circle version)

Problem. Three uniform random points on a circle. Probability that their triangle contains the centre?

Solution. The triangle contains the centre iff the points are not all in some semicircle. By the semicircle formula $P(\text{all in a semicircle}) = \frac{n}{2^{n-1}} = \frac{3}{4}$ for $n = 3$; answer $\frac{1}{4}$. Why the formula: the events “all points lie in the semicircle clockwise from point i ” are disjoint, each of probability $2^{-(n-1)}$. (The sphere with 4 points gives $\frac{1}{8}$, Putnam 1992 A6, via pairing each point with its antipode.)



PUTNAM-SPECIFIC HEURISTICS

- **Guess the answer numerically.** Evaluate the first few terms, approximate integrals; recognise π , $\ln 2$, e , $\sqrt{3}$.
- **Specific year numbers** (“2025”) are rarely special: solve for general n , then substitute. Check parity or residues of the year when relevant.
- **When a problem mixes areas** (a limit of a combinatorial quantity), each piece is usually standard; decompose.
- **Know the small toolkit cold:** Stolz, King’s rule, C-S, AM-GM, pigeonhole, determinants mod 2, linearity of expectation, generating functions, telescoping.

VII FURTHER WORKED EXAMPLES

one more pass through every area, plus how to write it up

ANATOMY OF A FULL-MARKS WRITE-UP

1. **Answer.** “The answer is $n = 3$.” (for “find all” problems)
2. **Verification.** Show the answer works.
3. **Setup.** Notation, WLOG with justification, reduction steps.
4. **Claims.** Each lemma stated, then proved. Label them (Claim 1, 2, ...).
5. **Conclusion.** Tie the claims together; address equality / boundary / small cases explicitly.

Graders reward *clearly stated* partial progress. A claim with a correct proof earns points even if the rest fails; an unlabelled page of computation earns nothing. Put the most important claim in a box or on its own line.

INVARIANTS & MONOVARIANTS

WORKED EXAMPLE · A quadratic monovariant IMO 1986/3

Problem. Integers $x_1, \dots, x_5$ sit on a pentagon with positive sum S . If $x_2 < 0$ (say) we may replace (x_1, x_2, x_3) by $(x_1 + x_2, -x_2, x_3 + x_2)$. Show the process always stops.

Solution. The sum S is invariant. Let $f = \sum_i (x_i - x_{i+2})^2$ (indices mod 5). A direct computation shows the move changes f by $2x_2S < 0$. So f is a nonnegative integer that strictly decreases: the process terminates.

Lesson. when the obvious quantities are invariant, try a quadratic form in differences.

ALGEBRA

WORKED EXAMPLE · Chebyshev after logs classical

Problem. $a, b, c > 0$. Prove $a^a b^b c^c \geq (abc)^{\frac{a+b+c}{3}}$.

Solution. Take logs: $\sum a \ln a \geq \frac{1}{3}(\sum a)(\sum \ln a)$. The sequences (a, b, c) and $(\ln a, \ln b, \ln c)$ are similarly ordered, so this is Chebyshev’s inequality.

WORKED EXAMPLE · A clever comparison IMO 2005/3

Problem. $x, y, z > 0$, $xyz \geq 1$. Prove $\sum \frac{x^5 - x^2}{x^5 + y^2 + z^2} \geq 0$.

Solution. Let $S = x^2 + y^2 + z^2$. Claim: $\frac{x^5 - x^2}{x^5 + y^2 + z^2} \geq \frac{x^5 - x^2}{x^3 S}$. The difference is $\frac{x^2(x^3 - 1)^2(y^2 + z^2)}{x^3 S(x^5 + y^2 + z^2)} \geq 0$. Now $\frac{x^5 - x^2}{x^3 S} = \frac{x^2 - \frac{1}{x}}{S} \geq \frac{x^2 - yz}{S}$ since $\frac{1}{x} \leq yz$. Summing, $\sum \frac{x^2 - yz}{S} = \frac{S - (xy + yz + zx)}{S} \geq 0$.

Lesson. replace a hard denominator by a common one, controlling the error by a perfect square.

WORKED EXAMPLE · Additive + multiplicative classical

Problem. Find all $f : \mathbb{R} \rightarrow \mathbb{R}$, not identically 0, with $f(x + y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$.

Solution. $f(x^2) = f(x)^2 \geq 0$, so $f \geq 0$ on $[0, \infty)$; with additivity f is increasing, hence $f(x) = cx$ (Cauchy). Then $c^2 = c$, $c = 1$: $f(x) = x$.

Lesson. a squared value gives a sign, a sign gives monotonicity, monotonicity tames Cauchy.

NUMBER THEORY

WORKED EXAMPLE · Pell equation classical

Problem. For which n is the triangular number $\frac{n(n+1)}{2}$ a perfect square m^2 ?

Solution. Multiply by 8: $(2n + 1)^2 - 8m^2 = 1$, a Pell equation $u^2 - 8m^2 = 1$ with fundamental solution $(3, 1)$. All solutions: $u_k + m_k\sqrt{8} = (3 + \sqrt{8})^k$, giving $(u, m) = (3, 1), (17, 6), (99, 35), (577, 204), \dots$ and $n = \frac{u-1}{2} = 1, 8, 49, 288, \dots$ (Ex. $\frac{49 \cdot 50}{2} = 35^2$).

Lesson. complete the square to reach $u^2 - 8m^2 = 1$.

WORKED EXAMPLE · Primes $\equiv 1 \pmod{4}$ classical

Problem. Show there are infinitely many primes $p \equiv 1 \pmod{4}$.

Solution. If $p_1, \dots, p_k$ were all, let $N = (2p_1 \cdots p_k)^2 + 1$ and q a prime factor. q is odd, $q \neq p_i$, and -1 is a square mod q , so $q \equiv 1 \pmod{4}$. Contradiction.

COMBINATORICS

WORKED EXAMPLE · Linear algebra over $\mathbb{F}_2$ Oddtown

Problem. Subsets $A_1, \dots, A_m \subseteq [n]$ have odd sizes and pairwise even intersections. Show $m \leq n$.

Solution. Let $v_i \in \mathbb{F}_2^n$ be indicator vectors: $v_i \cdot v_i = 1$, $v_i \cdot v_j = 0$. If $\sum c_i v_i = 0$, dot with v_j : $c_j = 0$. So the v_i are independent in an n -dimensional space: $m \leq n$.

WORKED EXAMPLE · Burnside classical

Problem. How many necklaces of 6 beads in 2 colours, up to rotation?

Solution. Rotation by k has $\gcd(k, 6)$ cycles, fixing $2^{\gcd(k, 6)}$ colourings. Average over $k = 0, \dots, 5$: $\frac{64 + 2 + 4 + 8 + 4 + 2}{6} = 14$.

WORKED EXAMPLE · Generating functions Euler

Problem. Partitions of n into distinct parts are equinumerous with partitions into odd parts.

Solution. $\prod_{k \geq 1} (1 + x^k) = \prod_k \frac{1 - x^{2k}}{1 - x^k} = \prod_{k \text{ odd}} \frac{1}{1 - x^k}$: the even factors cancel.

WORKED EXAMPLE · Euler’s formula classical

Problem. A simple planar graph with $V \geq 3$ has $E \leq 3V - 6$.

Solution. Each face is bounded by ≥ 3 edges and each edge borders ≤ 2 faces: $2E \geq 3F$. With $V - E + F = 2$: $2 = V - E + F \leq V - E + \frac{2E}{3}$, so $E \leq 3V - 6$.

WORKED EXAMPLE · Induction by insertion classical

Problem. Every tournament has a Hamiltonian path.

Solution. Induct. Remove v ; take a path $u_1 \rightarrow \dots \rightarrow u_{n-1}$. If $v \rightarrow u_1$ or $u_{n-1} \rightarrow v$, attach at an end. Otherwise $u_1 \rightarrow v$ and $v \rightarrow u_{n-1}$, so there is a first i with $v \rightarrow u_{i+1}$; then $u_i \rightarrow v$, and insert v between u_i and u_{i+1} .

Lesson. discrete continuity again: the arrow direction must switch somewhere.

GEOMETRY

WORKED EXAMPLE · Directed angles	Simson line
Problem. P on (ABC) has feet D, E, F on BC, CA, AB. Show D, E, F collinear. Solution. P, D, C, E are concyclic (right angles at D, E) and so are P, D, B, F. Hence $\angle(DE, DP) = \angle(CE, CP) = \angle(CA, CP)$ and $\angle(DF, DP) = \angle(BF, BP) = \angle(BA, BP)$. These are equal iff A, B, C, P are concyclic, which holds. So DE and DF are the same line. Lesson. directed angles avoid all configuration cases; the converse comes for free.	

WORKED EXAMPLE · Homothety	diameter of the incircle lemma
Problem. Incircle touches BC at D; X is the antipode of D. Show AX passes through the A-excircle's touch point D' on BC. Solution. The tangent to the incircle at X is parallel to BC. The homothety at A taking the incircle to the A-excircle maps this tangent line to the parallel tangent of the excircle on the same side, namely BC, and so maps X to the tangency point D'. Hence A, X, D' are collinear.	

WORKED EXAMPLE · Ptolemy	van Schooten
Problem. ABC equilateral, P on arc BC not containing A. Show $PA = PB + PC$. Solution. Ptolemy on $ABPC$: $PA \cdot BC = PB \cdot AC + PC \cdot AB$. Divide by the common side length.	

PUTNAM

WORKED EXAMPLE · Concentration of mass	classical
Problem. f continuous on $[0, 1]$. Show $n \int_0^1 x^n f(x) dx \rightarrow f(1)$. Solution. $n \int_0^1 x^n dx = \frac{n}{n+1} \rightarrow 1$, so it suffices to show $n \int_0^1 x^n (f(x) - f(1)) dx \rightarrow 0$. Given $\varepsilon > 0$ pick δ with $f(x) - f(1) < \varepsilon$ on $[1 - \delta, 1]$. That part contributes $< \varepsilon$; the rest is $\leq 2\ f\ _\infty n(1 - \delta)^{n+1} \rightarrow 0$. Lesson. split into “near the peak” and “far away”; the far part decays geometrically.	

WORKED EXAMPLE · Pairing in a group	classical
Problem. A finite group of even order has an element of order 2. Solution. Pair each g with g^{-1}. Elements with $g = g^{-1}$ are e and the elements of order 2. Non-self-paired elements come in pairs, so the number of self-paired elements is even; it includes e, so there is at least one more.	

WORKED EXAMPLE · First-step analysis	classical
Problem. Expected number of fair coin flips until HH? Until HT? Solution. For HH: states “start” (E_0) and “just saw H” (E_1): $E_0 = 1 + \frac{1}{2}E_1 + \frac{1}{2}E_0$, $E_1 = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2}E_0$. Solving: $E_0 = 6$. For HT: once you have an H, you wait for the first T (mean 2); total $2 + 2 = 4$. Lesson. patterns that overlap themselves (HH) take longer; set up equations over states.	

WORKED EXAMPLE · Newton's identities	classical
Problem. Complex $x_1, \dots, x_n$ and $y_1, \dots, y_n$ satisfy $\sum x_i^k = \sum y_i^k$ for $k = 1, \dots, n$. Show the multisets agree. Solution. Newton's identities determine $e_1, \dots, e_n$ from $p_1, \dots, p_n$ (in characteristic 0). So both sets have the same elementary symmetric functions, hence are the roots of the same monic polynomial.	

VIII REFERENCE TABLES

identities and numbers to have at your fingertips

SUMS & PRODUCTS

Finite sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \sum k^2 = \frac{n(n+1)(2n+1)}{6}, \sum k^3 = \left(\frac{n(n+1)}{2}\right)^2, \sum_{k=0}^{n-1} r^k = \frac{r^n - 1}{r - 1}, \sum_{k=1}^n kr^k = \frac{r - (n+1)r^{n+1} + nr^{n+2}}{(1-r)^2}. \text{ Falling powers: } \sum_{k=0}^{n-1} k^{\underline{m}} = \frac{n^{\underline{m+1}}}{m+1} \text{ (discrete calculus).}$$

Trigonometric

$$\sum_{k=1}^n \cos kx = \frac{\sin \frac{nx}{2} \cos \frac{(n+1)x}{2}}{\sin \frac{x}{2}}, \sum_{k=1}^n \sin kx = \frac{\sin \frac{nx}{2} \sin \frac{(n+1)x}{2}}{\sin \frac{x}{2}}.$$

$$\prod_{k=1}^{n-1} \sin \frac{k\pi}{n} = \frac{n}{2^{n-1}}; \sum_{k=1}^{n-1} \csc^2 \frac{k\pi}{n} = \frac{n^2-1}{3}; \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} = -\frac{1}{8};$$

$$\cos \frac{\pi}{5} = \frac{1+\sqrt{5}}{4}.$$

Product-to-sum: $2 \cos a \cos b = \cos(a-b) + \cos(a+b),$
 $2 \sin a \sin b = \cos(a-b) - \cos(a+b).$
 In a triangle: $\tan A + \tan B + \tan C = \tan A \tan B \tan C;$
 $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C; \cos^2 A + \cos^2 B + \cos^2 C + 2 \cos A \cos B \cos C = 1.$

Complex & roots of unity

$$\prod_{k=1}^{n-1} (1 - \omega^k) = n; \sum_{k=0}^{n-1} \omega^{jk} = n[n \mid j]; x^n - 1 = \prod (x - \omega^k);$$

$$|1 - e^{i\theta}| = 2|\sin \frac{\theta}{2}|; \text{ de Moivre } (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

(expand to get $\cos n\theta$ as a polynomial in $\cos \theta$).

SMALL NUMBERS

n	0	1	2	3	4	5	6	7	8	9	10
$n!$	1	1	2	6	24	120	720	5040	40320	$\approx 3.6e5$	$\approx 3.6e6$
F_n	0	1	1	2	3	5	8	13	21	34	55
C_n	1	1	2	5	14	42	132	429	1430	4862	16796
B_n	1	1	2	5	15	52	203	877	4140	21147	$\approx 1.2e5$
D_n	1	0	1	2	9	44	265	1854	14833	$\approx 1.3e5$	$\approx 1.3e6$
$p(n)$	1	1	2	3	5	7	11	15	22	30	42
2^n	1	2	4	8	16	32	64	128	256	512	1024

Primes < 100 : 2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97 (25 of them).

Useful factorizations: $1001 = 7 \cdot 11 \cdot 13$; $2^{10} - 1 = 3 \cdot 11 \cdot 31$; $2^{11} - 1 = 23 \cdot 89$; $2^{32} + 1 = 641 \cdot 6700417$; $10^6 - 1 = 3^3 \cdot 7 \cdot 11 \cdot 13 \cdot 37$; $2025 = 45^2 = 3^4 \cdot 5^2$; $2026 = 2 \cdot 1013$ (1013 prime); 2027 prime.

Perfect powers near each other: $2^3 + 1 = 3^2$ is the only consecutive pair (Mihăilescu).

MOD TABLES

modulus	k -th powers attain	use for
3	squares 0, 1	sum of squares arguments
4	squares 0, 1	$x^2 + y^2 \neq 4k + 3$
8	odd squares $\equiv 1$	$x^2 + y^2 + z^2 \neq 8k + 7$
9	cubes 0, ± 1	sums of cubes
7	cubes 0, ± 1	sums of cubes
16	4th powers 0, 1	$\sum x_i^4$ problems
11	5th powers 0, ± 1	x^5 equations
13	4th powers 0, 1, 3, 9; cubes 0, $\pm 1, \pm 5$	mixed

INEQUALITY EQUALITY-CASE CHEAT TABLE

equality at	suspects
$a = b = c$ only	AM-GM, C-S, Jensen, power mean, SOS
$a = b = c$ and $(t, t, 0)$	Schur, degree-4 Schur, uvw
boundary $c \rightarrow 0$ only	mixing variables, SOS with $S_c \geq 0$
$a = b \neq c$	uvw with $b = c$ check, tangent line fails
no equality (strict)	weaker tool suffices; look for slack

EXAM-DAY CHECKLIST

1. Read all problems. Rank by how quickly you can *see* an approach, not by number.
2. Spend the first 20–30 minutes per problem on examples and a conjecture; don't start writing prematurely.
3. When you have a solution, write it *immediately* and completely; then move on.
4. Every 45 minutes, re-evaluate: is this still the best use of the next 45?
5. On a hard problem, record partial results as claims: lemmas, bounds, small cases, the answer. IMO graders award these.
6. Final 15 minutes: reread write-ups for gaps (equality cases, n small, "WLOG" justified, both directions of "iff").

- Strategy
- Algebra
- Number theory
- Combinatorics
- Geometry
- Putnam analysis

Problem solving is mostly asking the right small question first.

IX FORMULA COMPENDIUM

everything worth having memorised, in one place

ALGEBRAIC IDENTITIES

Factorisations

$a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$; $a^n + b^n = (a + b)(a^{n-1} - a^{n-2}b + \dots + b^{n-1})$ (n odd).
 $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$.
 $(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b)(b + c)(c + a)$.
 $a^4 + 4b^4 = (a^2 + 2ab + 2b^2)(a^2 - 2ab + 2b^2)$ (Sophie Germain).
 $a^4 + a^2 + 1 = (a^2 + a + 1)(a^2 - a + 1)$; $x^5 + x + 1 = (x^2 + x + 1)(x^3 - x^2 + 1)$.
 $(a - b)^3 + (b - c)^3 + (c - a)^3 = 3(a - b)(b - c)(c - a)$.
 $a^2(b - c) + b^2(c - a) + c^2(a - b) = -(a - b)(b - c)(c - a)$.
 $(a - 1)(b - 1)(c - 1) = abc - (ab + bc + ca) + (a + b + c) - 1$ (useful when $abc = 1$ or $\sum a = \sum ab$).

Sums of squares identities

Lagrange: $(\sum a_i^2)(\sum b_i^2) - (\sum a_i b_i)^2 = \sum_{i < j} (a_i b_j - a_j b_i)^2$.
Brahmagupta–Fibonacci: $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$.
Euler four-square: products of sums of four squares are sums of four squares.
 $\sum_{\text{cyc}} (a - b)^2 = 2(\sum a^2 - \sum ab)$.

Three variables in $p = \sum a, q = \sum ab, r = abc$

$\sum a^2 = p^2 - 2q$; $\sum a^3 = p^3 - 3pq + 3r$; $\sum a^4 = p^4 - 4p^2q + 2q^2 + 4pr$.
 $\sum a^2 b^2 = q^2 - 2pr$; $\sum_{\text{sym}} a^2 b = pq - 3r$; $(a + b)(b + c)(c + a) = pq - r$.
 $\sum \frac{1}{a} = \frac{q}{r}$; $(1 + a)(1 + b)(1 + c) = 1 + p + q + r$; $\prod (a - b)^2 = p^2 q^2 - 4q^3 - 4p^3 r + 18pqr - 27r^2$.

Binomial & multinomial

$(x + y)^n = \sum \binom{n}{k} x^k y^{n-k}$; $(x_1 + \dots + x_m)^n = \sum \frac{n!}{k_1! \dots k_m!} \prod x_i^{k_i}$.
Generalised: $(1 + x)^\alpha = \sum_{k \geq 0} \binom{\alpha}{k} x^k$, $|x| < 1$, $\binom{\alpha}{k} = \frac{\alpha(\alpha-1)\dots(\alpha-k+1)}{k!}$.
 $\binom{-n}{k} = (-1)^k \binom{n+k-1}{k}$; $\binom{1/2}{k} = \frac{(-1)^{k-1}}{4^k(2k-1)} \binom{2k}{k}$.

MORE INEQUALITIES

Named inequalities

Nesbitt: $\sum \frac{a}{b+c} \geq \frac{3}{2}$. **Minkowski:** $\sqrt{\sum (a_i + b_i)^2} \leq \sqrt{\sum a_i^2} + \sqrt{\sum b_i^2}$; generally $\|x + y\|_p \leq \|x\|_p + \|y\|_p$ ($p \geq 1$).
Maclaurin: with $S_k = e_k / \binom{n}{k}$ ($x_i \geq 0$): $S_1 \geq S_2^{1/2} \geq \dots \geq S_n^{1/n}$.
Newton: $S_k^2 \geq S_{k-1} S_{k+1}$ (real x_i).
Schur (degree 4): $\sum a^2(a - b)(a - c) \geq 0$.
Huygens: $\prod (1 + a_i) \geq (1 + \sqrt[n]{\prod a_i})^n$.
Aczél: $(a_1^2 - \sum_{i \geq 2} a_i^2)(b_1^2 - \sum_{i \geq 2} b_i^2) \leq (a_1 b_1 - \sum_{i \geq 2} a_i b_i)^2$ if $a_1^2 > \sum_{i \geq 2} a_i^2$.
Radon: $\sum \frac{x_i^{p+1}}{a_i^p} \geq \frac{(\sum x_i)^{p+1}}{(\sum a_i)^p}$ ($p > 0$).

Elementary bounds

$e^x \geq 1 + x$; $\frac{x}{1+x} \leq \ln(1+x) \leq x$ ($x > -1$); $\sin x \leq x \leq \tan x$ on $[0, \frac{\pi}{2})$; **Jordan** $\sin x \geq \frac{2x}{\pi}$ on $[0, \frac{\pi}{2}]$; $\cos x \geq 1 - \frac{x^2}{2}$; $(1 + \frac{1}{n})^n < e < (1 + \frac{1}{n})^{n+1}$; $2\sqrt{n+1} - 2 < \sum_{k=1}^n \frac{1}{\sqrt{k}} < 2\sqrt{n} - 1$ ($n \geq 2$); $\frac{4^n}{2\sqrt{n}} \leq \binom{2n}{n} \leq \frac{4^n}{\sqrt{3n+1}}$; $n! \geq (n/e)^n$.

Triangle inequalities

$R \geq 2r$ (Euler); $a^2 + b^2 + c^2 \geq 4\sqrt{3}\Delta$ (Weitzenböck); $s \geq 3\sqrt{3}r$; $abc \geq (a + b - c)(b + c - a)(c + a - b)$; $\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$; $\cos A + \cos B + \cos C \leq \frac{3}{2}$; $\cos A \cos B \cos C \leq \frac{1}{8}$; $a^2 + b^2 + c^2 \leq 9R^2$.
Erdős–Mordell: P inside, $PA + PB + PC \geq 2(PD + PE + PF)$ (distances to sides).

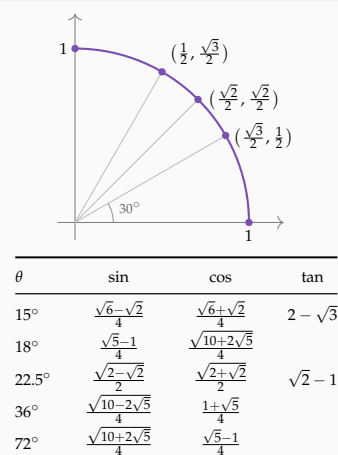
TRIGONOMETRY

Addition & multiple angles

$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$; $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$; $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$.
 $\sin 2x = 2 \sin x \cos x$; $\cos 2x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$; $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$.
 $\sin 3x = 3 \sin x - 4 \sin^3 x$; $\cos 3x = 4 \cos^3 x - 3 \cos x$; $\tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$.
Half-angle: $\sin^2 \frac{x}{2} = \frac{1 - \cos x}{2}$, $\tan \frac{x}{2} = \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$.
Weierstrass $t = \tan \frac{x}{2}$: $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $dx = \frac{2dt}{1+t^2}$.

Sum $\leftrightarrow$ product

$\sin a + \sin b = 2 \sin \frac{a+b}{2} \cos \frac{a-b}{2}$; $\sin a - \sin b = 2 \cos \frac{a+b}{2} \sin \frac{a-b}{2}$;
 $\cos a + \cos b = 2 \cos \frac{a+b}{2} \cos \frac{a-b}{2}$; $\cos a - \cos b = -2 \sin \frac{a+b}{2} \sin \frac{a-b}{2}$;
 $a \sin x + b \cos x = \sqrt{a^2 + b^2} \sin(x + \varphi)$, $\tan \varphi = \frac{b}{a}$.



Identities in a triangle ($A + B + C = \pi$)

$\sum \tan A = \prod \tan A$; $\sum \cot A \cot B = 1$; $\sum \tan \frac{A}{2} \tan \frac{B}{2} = 1$.
 $\sum \sin 2A = 4 \prod \sin A$; $\sum \cos 2A = -1 - 4 \prod \cos A$; $\sum \cos^2 A = 1 - 2 \prod \cos A$.
 $\sum \sin A = 4 \prod \cos \frac{A}{2}$; $\sum \cos A = 1 + 4 \prod \sin \frac{A}{2} = 1 + \frac{r}{R}$.
Projection: $a = b \cos C + c \cos B$. **Tangents:** $\frac{a-b}{a+b} = \frac{\tan \frac{A-B}{2}}{\tan \frac{A+B}{2}}$.

NUMBER-THEORETIC FORMULAS

Arithmetic functions

$\varphi(p^k) = p^{k-1}(p-1)$; $\sum_{d|n} \mu(d) = [n=1]$; $\sum_{d|n} \varphi(d) = n$; $\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d}$; $\sum_{k \leq n, \gcd(k,n)=1} k = \frac{n\varphi(n)}{2}$ ($n > 1$).
 $\sum_{k=1}^n \tau(k) = \sum_{k=1}^n \lfloor \frac{n}{k} \rfloor$; $\sum_{k=1}^n \sigma(k) = \sum_k k \lfloor \frac{n}{k} \rfloor$; $\tau(n) < 2\sqrt{n}$; $\prod_{d|n} d = n^{\tau(n)/2}$.
 Density of coprime pairs $= \frac{6}{\pi^2}$; $\sum_{k \leq n} \varphi(k) \sim \frac{3n^2}{\pi^2}$.
 Even perfect numbers $= 2^{p-1}(2^p - 1)$ with $2^p - 1$ prime (Euclid–Euler).

gcd identities

$\gcd(a, b) \operatorname{lcm}(a, b) = ab$; $\gcd(a^m - b^m, a^n - b^n) = a^{\gcd(m, n)} - b^{\gcd(m, n)}$
 $(\gcd(a, b) = 1)$; $\gcd(F_m, F_n) = F_{\gcd(m, n)}$; $\gcd(n, n+1) = 1$;
 $\gcd(a, b) = \gcd(a, b - ka)$; $\gcd(2^a + 1, 2^b + 1) = 2^s + 1$ if $a/g, b/g$ both odd, else 1 ($g = \gcd(a, b)$).

Orders & structure

$(\mathbb{Z}/2^k)^\times \cong C_2 \times C_{2^{k-2}}$ ($k \geq 3$), generated by $-1, 5$; $\operatorname{ord}_{2^k}(5) = 2^{k-2}$.
 Carmichael $\lambda(n)$: exponent of $(\mathbb{Z}/n)^\times$; $\lambda(p^k) = \varphi(p^k)$ for odd p ,
 $\lambda(2^k) = 2^{k-2}$ ($k \geq 3$), $\lambda(\operatorname{lcm}) = \operatorname{lcm}(\lambda)$.
 If $\operatorname{ord}_p(a) = d$ and $p^e \parallel a^d - 1$ (p odd) then $\operatorname{ord}_{p^{e+j}}(a) = dp^j$.
 $x^2 \equiv a \pmod{p}$ (p odd) has $1 + \left(\frac{a}{p}\right)$ solutions; $x^2 \equiv 1 \pmod{2^k}$ has 4 solutions for $k \geq 3$.

Representations

$n = \sum$ of two squares in $r_2(n) = 4 \sum_{d|n} \chi(d)$ ways ($\chi(d) = \pm 1$ for $d \equiv \pm 1 \pmod{4}$, 0 if even). Pell: $x^2 - dy^2 = -1$ solvable iff the continued fraction period of $\sqrt{d}$ is odd. Continued fractions: $p_n = a_n p_{n-1} + p_{n-2}$, $q_n = a_n q_{n-1} + q_{n-2}$, $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$.

GEOMETRY FORMULAS

Areas

Triangle: $\Delta = \frac{1}{2}ah_a = \frac{1}{2}bc \sin A = rs = \frac{abc}{4R} = 2R^2 \sin A \sin B \sin C = \sqrt{r r_a r_b r_c}$.
 Shoelace: $\Delta = \frac{1}{2} |\sum_i (x_i y_{i+1} - x_{i+1} y_i)|$; vectors: $\Delta = \frac{1}{2} |\vec{u} \times \vec{v}|$.
 Cyclic quadrilateral (Brahmagupta): $\sqrt{(s-a)(s-b)(s-c)(s-d)}$;
 general (Bretschneider): subtract $abcd \cos^2 \frac{A+C}{2}$ under the root.
 Quadrilateral with diagonals p, q at angle θ : $\frac{1}{2}pq \sin \theta$. Regular n -gon, circumradius R : $\frac{n}{2}R^2 \sin \frac{2\pi}{n}$.

Lengths in a triangle

Bisector: $t_a^2 = bc \left(1 - \frac{a^2}{(b+c)^2}\right)$, $t_a = \frac{2bc \cos \frac{A}{2}}{b+c}$. Altitude $h_a = \frac{2\Delta}{a} = b \sin C$.
 $AI = \frac{r}{\sin \frac{A}{2}} = 4R \sin \frac{B}{2} \sin \frac{C}{2}$; $\tan \frac{A}{2} = \frac{r}{s-a}$; $r_a = s \tan \frac{A}{2}$.
 $r_a + r_b + r_c = 4R + r$; $\frac{1}{r} = \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c}$; right triangle: $r = \frac{a+b-c}{2}$ (c hypotenuse).
 $OG^2 = R^2 - \frac{a^2+b^2+c^2}{9}$; $GA^2 + GB^2 + GC^2 = \frac{a^2+b^2+c^2}{3}$.
Leibniz: $PA^2 + PB^2 + PC^2 = 3PG^2 + GA^2 + GB^2 + GC^2$.
 Cyclic quadrilateral: $R = \frac{1}{4K} \sqrt{(ab+cd)(ac+bd)(ad+bc)}$ (K its area).

COMBINATORIAL FORMULAS

Binomial identities

Pascal $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$; absorption $k \binom{n}{k} = n \binom{n-1}{k-1}$; $\binom{n}{k} \binom{k}{m} = \binom{n}{m} \binom{n-m}{k-m}$;
 $\sum_k (-1)^k \binom{n}{k} = 0$; $\sum_k \binom{n}{2k} = 2^{n-1}$; $\sum_k \binom{n-k}{k} = F_{n+1}$; $\sum_k k^2 \binom{n}{k} = n(n+1)2^{n-2}$;
 $\sum_{k=0}^m (-1)^k \binom{n}{m-k} = (-1)^m \binom{n-1}{m-1}$; $\sum_k \binom{2k}{k} \binom{2n-2k}{n-k} = 4^n$; $\sum_{k=0}^n \binom{k}{m} \binom{n-k}{r} = \binom{n+1}{m+r+1}$ (Chu).
 $\sum_{k \geq 1} \frac{(-1)^{k-1}}{k} \binom{n}{k} = H_n$; $\binom{2n}{n} = \sum_k \binom{n}{k}^2$; $\sum_k \binom{n}{k} \binom{m}{r+k} = \binom{n+m}{n+r}$.

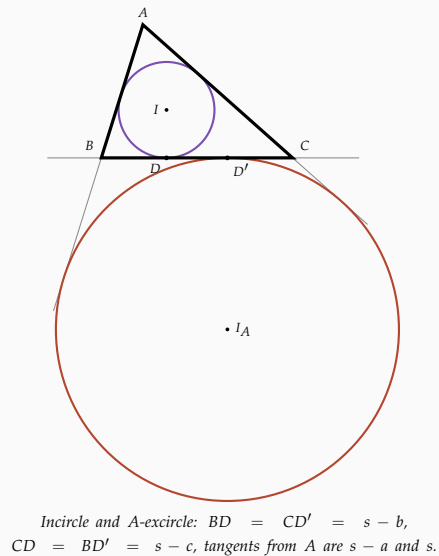
Counting formulas

Derangements $D_n = (n-1)(D_{n-1} + D_{n-2}) = nD_{n-1} + (-1)^n$. Stirling: $S(n, 2) = 2^{n-1} - 1$, $S(n, n-1) = \binom{n}{2}$, $c(n, 1) = (n-1)!$;
 $x^n = \sum_k S(n, k) x^k$.
 Ballot: candidate A with a votes stays strictly ahead of B with $b < a$ throughout w.p. $\frac{a-b}{a+b}$.
 Paths $(0, 0) \rightarrow (m, n)$: $\binom{m+n}{m}$; not touching the diagonal (for $m > n$): $\frac{m-n}{m+n} \binom{m+n}{m}$.
 Regions: n lines in general position: $1 + n + \binom{n}{2}$; chords among n points on a circle (general position): $1 + \binom{n}{2} + \binom{n}{4}$; n planes in space: $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3}$.
 Spanning trees: K_n : n^{n-2} ; $K_{m,n}$: $m^{n-1} n^{m-1}$; any graph: Kirchhoff's matrix-tree theorem (any cofactor of the Laplacian).
 Labelled graphs on n vertices: $2^{\binom{n}{2}}$. Triangulations of convex $(n+2)$ -gon: C_n .
 Pentagonal theorem: $\prod_{k \geq 1} (1 - x^k) = \sum_{k \in \mathbb{Z}} (-1)^k x^{k(3k-1)/2}$.

Probability formulas

$E[X] = \sum P(X \geq k)$ ($X \in \mathbb{N}$); $\operatorname{Var}(X) = E[X^2] - E[X]^2$; $\operatorname{Var}(\sum X_i) = \sum \operatorname{Var} X_i + 2 \sum_{i < j} \operatorname{Cov}(X_i, X_j)$.
 Markov $P(X \geq a) \leq \frac{EX}{a}$; Chebyshev $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$; union bound; Bayes $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$.

Binomial(n, p)	mean np	var $np(1-p)$
Geometric(p) on $\{1, 2, \dots\}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson(λ)	λ	λ
Uniform($[a, b]$)	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ)	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal(μ, σ^2)	μ	σ^2
Hypergeometric	$\frac{nK}{N}$	



Coordinates & vectors

Distance from (x_0, y_0) to $ax + by + c = 0$: $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$. Reflection of $\vec{p}$ in the line through $\vec{0}$ with unit direction $\vec{u}$: $2(\vec{p} \cdot \vec{u})\vec{u} - \vec{p}$.
 Rotation by θ : $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$. With O as origin: $\vec{H} = \vec{A} + \vec{B} + \vec{C}$, $\vec{G} = \frac{1}{3}\vec{H}$, $\vec{N} = \frac{1}{2}\vec{H}$.
 Tetrahedron volume $\frac{1}{6}|\det(\vec{b} - \vec{a}, \vec{c} - \vec{a}, \vec{d} - \vec{a})|$. Sphere $4\pi r^2$, $\frac{4}{3}\pi r^3$; cone $\frac{1}{3}\pi r^2 h$, lateral area $\pi r \ell$.

Complex-number formulas

Signed area of abc : $\frac{i}{4} \det \begin{pmatrix} a & a & 1 \\ b & b & 1 \\ c & c & 1 \end{pmatrix}$. Rotation of z about w : $w + (z - w)e^{i\theta}$.
 Directly similar $\triangle abc \sim \triangle a'b'c' \iff \frac{a-b}{a-c} = \frac{a'-b'}{a'-c'}$.
 Unit circle, chords ab and cd meet at $\frac{ab(c+d) - cd(a+b)}{ab - cd}$. Projection of z onto chord ab : $\frac{1}{2}(a + b + z - ab\bar{z})$.

CALCULUS & ANALYSIS FORMULAS

Derivatives

$(\tan x)' = \sec^2 x$; $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$; $(\arctan x)' = \frac{1}{1+x^2}$; $(a^x)' = a^x \ln a$; $(x^x)' = x^x(1 + \ln x)$.
 Leibniz: $(fg)^{(n)} = \sum \binom{n}{k} f^{(k)} g^{(n-k)}$. Inverse: $(f^{-1})'(y) = \frac{1}{f'(x)}$.
 Under the integral: $\frac{d}{dt} \int_{a(t)}^{b(t)} f(x, t) dx = \int_{a(t)}^{b(t)} f(x, t) b' - f(a, t) a' + \int_a^b \partial_t f dx$.

Antiderivatives

$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctan \frac{x}{a}$; $\int \frac{dx}{\sqrt{x^2 - a^2}} = \arcsin \frac{x}{a}$; $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right|$;
 $\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{x^2 + a^2})$; $\int \sec x = \ln |\sec x + \tan x|$; $\int \tan x = -\ln |\cos x|$;
 $\int \ln x = x \ln x - x$; $\int e^{ax} \sin bx = \frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2}$; $\int \sqrt{a^2 - x^2} = \frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \arcsin \frac{x}{a}$.
 Reduction: $\int_0^{\pi/2} \sin^n x dx = \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2} x dx$.

Taylor series at 0

$e^x = \sum \frac{x^n}{n!}$; $\sin x = \sum \frac{(-1)^n x^{2n+1}}{(2n+1)!}$; $\cos x = \sum \frac{(-1)^n x^{2n}}{(2n)!}$; $\ln(1+x) = \sum_{n \geq 1} \frac{(-1)^{n-1} x^n}{n}$;
 $\arctan x = \sum \frac{(-1)^n x^{2n+1}}{2n+1}$; $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$; $\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \dots$;
 $\frac{x}{e^x - 1} = \sum B_n \frac{x^n}{n!}$ with $B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$.

Standard limits

$\frac{\sin x}{x} \rightarrow 1$; $\frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}$; $\frac{e^x - 1}{x} \rightarrow 1$; $\frac{a^x - 1}{x} \rightarrow \ln a$; $\frac{\ln(1+x)}{x} \rightarrow 1$;
 $x \ln x \rightarrow 0^-$; $x^x \rightarrow 1$ ($x \rightarrow 0^+$); $n^{1/n} \rightarrow 1$; $(1 + \frac{a}{n})^n \rightarrow e^a$; $\frac{x^k}{e^x} \rightarrow 0$.

EQUATIONS & POLYNOMIALS

Low-degree equations

$ax^2 + bx + c = 0$: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Cubic: $x = t - \frac{b}{3a}$ gives $t^3 + pt + q = 0$, discriminant $\Delta = -4p^3 - 27q^2$ ($\Delta > 0$: three real roots), **Cardano** $t = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}$. Three real roots: $t = 2\sqrt{-\frac{p}{3}} \cos(\frac{1}{3} \arccos(\frac{3q}{2p}\sqrt{-\frac{3}{p}}) - \frac{2\pi k}{3})$.
 Quartic $x^4 + ax^3 + bx^2 + cx + d$: Vieta $\sum r_i = -a$, $\sum r_i r_j = b$, $\sum r_i r_j r_k = -c$, $\prod r_i = d$.

Special functions

$\Gamma(x+1) = x\Gamma(x)$, $\Gamma(n+1) = n!$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. $B(x, y) = \int_0^1 t^{x-1}(1-t)^{y-1} dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$.
 Reflection $\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x}$; duplication $\Gamma(x)\Gamma(x + \frac{1}{2}) = 2^{1-2x} \sqrt{\pi} \Gamma(2x)$.
 $\zeta(2n) = \frac{(-1)^{n+1} B_{2n} (2\pi)^{2n}}{2(2n)!}$: $\zeta(2) = \frac{\pi^2}{6}$, $\zeta(4) = \frac{\pi^4}{90}$, $\zeta(6) = \frac{\pi^6}{945}$.
 $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$; $\eta(s) = \sum \frac{(-1)^{n-1}}{n^s} = (1 - 2^{1-s})\zeta(s)$.

Euler–Maclaurin & Stirling

$\sum_{k=a}^b f(k) = \int_a^b f + \frac{f(a)+f(b)}{2} + \sum_{j \geq 1} \frac{B_{2j}}{(2j)!} (f^{(2j-1)}(b) - f^{(2j-1)}(a)) + R$.
 $\ln n! = n \ln n - n + \frac{1}{2} \ln(2\pi n) + \frac{1}{12n} - \frac{1}{360n^3} + \dots$.
 Faulhaber: $\sum_{k=1}^n k^p = \frac{1}{p+1} \sum_{j=0}^p \binom{p+1}{j} B_j^+ n^{p+1-j}$ ($B_1^+ = +\frac{1}{2}$).

Fourier & ODE

On $(-\pi, \pi)$: $x = 2 \sum_{n \geq 1} \frac{(-1)^{n+1} \sin nx}{n}$; $x^2 = \frac{\pi^2}{3} + 4 \sum_{n \geq 1} \frac{(-1)^n \cos nx}{n^2}$.
 Parseval: $\frac{1}{\pi} \int_{-\pi}^{\pi} f^2 = \frac{a_0^2}{2} + \sum (a_n^2 + b_n^2)$.
 $y' + p(x)y = q(x)$: $(ye^{\int p})' = qe^{\int p}$. $y'' + ay' + by = 0$ with roots $r_{1,2}$ of $r^2 + ar + b$: $c_1 e^{r_1 x} + c_2 e^{r_2 x}$; double root: $(c_1 + c_2 x)e^{rx}$; $\alpha \pm i\beta$: $e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$.
 Polar: $dx dy = r dr d\theta$. Green: area $= \frac{1}{2} \oint (x dy - y dx)$.

LINEAR & ABSTRACT ALGEBRA FORMULAS

Matrices

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$; eigenvalues of 2×2 : $\lambda^2 - (\text{tr } A)\lambda + \det A = 0$; $A^2 = (\text{tr } A)A - (\det A)I$.
 $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$. Cramer: $x_i = \frac{\det A_i}{\det A}$.
Sylvester: $\det(I_m + AB) = \det(I_n + BA)$. $\det e^A = e^{\text{tr } A}$.
Sherman–Morrison: $(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1 + v^T A^{-1}u}$.
Cauchy–Binet: $\det(AB) = \sum_S \det A_{[m],S} \det B_{S,[m]}$ (A : $m \times n$, B : $n \times m$).
 $\det(aI + bJ) = a^{n-1}(a + nb)$; J has eigenvalues n (once), 0 ($n-1$ times).
 Tridiagonal (diagonal a , off-diagonals b, c): $D_n = aD_{n-1} - bcD_{n-2}$.
Cauchy determinant: $\det \left[\frac{1}{x_i + y_j} \right] = \frac{\prod_{i < j} (x_j - x_i)(y_j - y_i)}{\prod_{i,j} (x_i + y_j)}$.

Groups & finite fields

Permutation order = lcm of cycle lengths; sign $= (-1)^{n - \# \text{cycles}}$, number with m_k cycles of length k : $\frac{n!}{\prod k^{m_k} m_k!}$.
 $|GL_n(\mathbb{F}_q)| = \prod_{k=0}^{n-1} (q^n - q^k)$. Monic irreducibles of degree n over $\mathbb{F}_q$: $\frac{1}{n} \sum_{d|n} \mu(d) q^{n/d}$.

CONSTANTS

π	3.14159265	e	2.71828183
γ	0.57721566	φ	1.61803399
$\ln 2$	0.69314718	$\ln 3$	1.09861229
$\ln 10$	2.30258509	$\log_{10} 2$	0.30103000
$\sqrt{2}$	1.41421356	$\sqrt{3}$	1.73205081
$\sqrt{5}$	2.23606798	$\log_{10} 3$	0.47712125
$\zeta(3)$	1.20205690	Catalan G	0.91596559
e^π	23.1406926	π^e	22.4591577
2^{10}	1024	2^{20}	1048576

Cyclotomic polynomials

Φ_1 $x - 1$ Φ_6 $x^2 - x + 1$
 Φ_2 $x + 1$ Φ_8 $x^4 + 1$
 Φ_3 $x^2 + x + 1$ Φ_9 $x^6 + x^3 + 1$
 Φ_4 $x^2 + 1$ Φ_{10} $x^4 - x^3 + x^2 - x + 1$
 Φ_5 $x^4 + x^3 + x^2 + x + 1$ Φ_{12} $x^4 - x^2 + 1$
 $\deg \Phi_n = \varphi(n)$; $\Phi_p(x) = \frac{x^p - 1}{x - 1}$; $\Phi_{p^k}(x) = \Phi_p(x^{p^{k-1}})$; $\Phi_{2n}(x) = \Phi_n(-x)$ ($n > 1$ odd); $\Phi_n(1) = p$ if $n = p^k$, else 1 (n not a prime power, $n > 1$); $\Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}$.

Roots-of-unity sums & products ($\omega = e^{2\pi i/n}$)

$$\prod_{k=1}^{n-1} (x - \omega^k) = 1 + x + \cdots + x^{n-1}; \quad \sum_{k=1}^{n-1} \frac{1}{1 - \omega^k} = \frac{n-1}{2};$$

$$\sum_{k=0}^{n-1} \cos \frac{2\pi k}{n} = 0 \quad (n \geq 2).$$

Gauss sum: $|\sum_{k=0}^{p-1} e^{2\pi i k^2/p}| = \sqrt{p}$ (p odd prime).

$$\sum_k \binom{n}{k} \cos k\theta = 2^n \cos \frac{\theta}{2} \cos \frac{n\theta}{2}; \quad \prod_{k=1}^n \cos \frac{k\pi}{2n+1} = \frac{1}{2^n};$$

$$\sum_{k=1}^n \cot^2 \frac{k\pi}{2n+1} = \frac{n(2n-1)}{3} \quad (\text{gives } \zeta(2) \text{ by squeezing}).$$

SUMS, SEQUENCES, FLOORS

Closed forms

$$\sum k^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}; \quad \sum k(k+1) = \frac{n(n+1)(n+2)}{3};$$

$$\sum_{k=1}^n \frac{1}{k(k+1)(k+2)} = \frac{1}{4} - \frac{1}{2(n+1)(n+2)};$$

$$\sum_{k \leq n} k \cdot k! = (n+1)! - 1; \quad \sum_{k \leq n} H_k = (n+1)H_n - n; \quad \sum_{k \leq n} \binom{k}{m} = \binom{n+1}{m+1}.$$

Fibonacci & Lucas ($L_0 = 2, L_1 = 1$)

$$L_n = F_{n-1} + F_{n+1} = \varphi^n + \psi^n; \quad F_{2n} = F_n L_n; \quad \sum_{k \leq n} F_k^2 = F_n F_{n+1};$$

$$\sum_{k \leq n} F_{2k-1} = F_{2n};$$

d'Ocagne $F_m F_{n+1} - F_{m+1} F_n = (-1)^n F_{m-n}; \quad F_m \mid F_n \iff m \mid n$ ($m \geq 3$); $L_n^2 - 5F_n^2 = 4(-1)^n$.

Lucas: 2, 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, 199, 322.

Floor & fractional part

$$\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x+y \rfloor \leq \lfloor x \rfloor + \lfloor y \rfloor + 1; \quad \lfloor \frac{\lfloor x \rfloor}{n} \rfloor = \lfloor \frac{x}{n} \rfloor; \quad \lceil x \rceil = -\lfloor -x \rfloor;$$

$$\lfloor x \rfloor + \lfloor -x \rfloor = -1 \quad (x \notin \mathbb{Z}).$$

$\#\{k \leq n : d \mid k\} = \lfloor \frac{n}{d} \rfloor; \quad \lfloor \frac{n}{k} \rfloor$ takes at most $2\sqrt{n}$ values (group sums by value).

$$\{x\} + \{y\} \geq \{x+y\}; \quad \lfloor 2x \rfloor + \lfloor 2y \rfloor \geq \lfloor x \rfloor + \lfloor y \rfloor + \lfloor x+y \rfloor \quad (\text{proves } \frac{(2m)!(2n)!}{m!n!(m+n)!} \in \mathbb{Z}).$$

INEQUALITY LIBRARY

Basic three-variable facts ($a, b, c \geq 0$)

$$\sum a^2 \geq \sum ab; \quad (\sum ab)^2 \geq 3abc \sum a; \quad \sum a^2 b^2 \geq abc \sum a; \quad (\sum a)^3 \geq 27abc;$$

$$(a+b)(b+c)(c+a) \geq 8abc; \quad (a+b)(b+c)(c+a) \geq \frac{8}{9}(a+b+c)(ab+bc+ca);$$

$$a^3 + b^3 \geq ab(a+b); \quad \sum \frac{a^2}{b} \geq \sum a; \quad \sum \frac{a}{b} \geq 3; \quad \sum \frac{1}{a+b} \geq \frac{9}{2\sum a}.$$

Deeper inequalities

Popoviciu (f convex): $f(x) + f(y) + f(z) + 3f(\frac{x+y+z}{3}) \geq 2[f(\frac{x+y}{2}) + f(\frac{y+z}{2}) + f(\frac{z+x}{2})]$.

Hardy ($p > 1, a_n \geq 0, A_n = a_1 + \cdots + a_n$): $\sum (\frac{A_n}{n})^p \leq (\frac{p}{p-1})^p \sum a_n^p$.

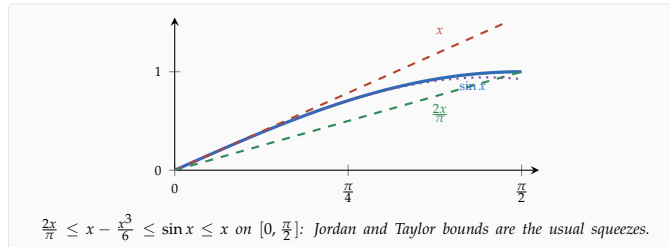
Carleman: $\sum (a_1 \cdots a_n)^{1/n} \leq e \sum a_n$.

Wirtinger: $f(0) = f(\pi) = 0 \Rightarrow \int_0^\pi f'^2 \geq \int_0^\pi f^2$.

Grönwall: $u' \leq \beta u \Rightarrow u(t) \leq u(0)e^{\beta t}$.

Landau: on $\mathbb{R}, \|f'\|_\infty^2 \leq 2\|f\|_\infty \|f''\|_\infty$ (constant 4 on a half-line).

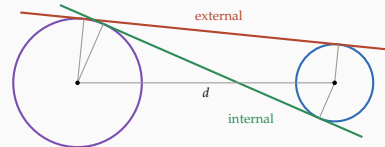
Chebyshev sum (opposite order) $n \sum a_i b_i \leq \sum a_i \sum b_i$.



MORE GEOMETRY

Circles

Tangent length from distance d : $\sqrt{d^2 - r^2}$. Common tangents of circles r_1, r_2 at distance d : external $\sqrt{d^2 - (r_1 - r_2)^2}$, internal $\sqrt{d^2 - (r_1 + r_2)^2}$. Angle between chords meeting inside: half the sum of intercepted arcs; outside (secants/tangents): half the difference. Chord = $2R \sin \frac{\theta}{2}$ for central angle θ . **Descartes**: mutually tangent circles with curvatures k_i (– for internal tangency): $(k_1 + k_2 + k_3 + k_4)^2 = 2 \sum k_i^2$.



Distances between centres

$$OI^2 = R^2 - 2Rr; \quad OH^2 = 9R^2 - (a^2 + b^2 + c^2); \quad IH^2 = 2r^2 - 4R^2 \cos A \cos B \cos C; \quad ON = \frac{OH}{2}.$$

Distance from O to BC : $R \cos A$; from H to BC : $2R \cos B \cos C$; $AH = 2R \cos A$.

Powers w.r.t. (ABC) : $\text{Pow}(I) = -2Rr$, $\text{Pow}(G) = -\frac{a^2+b^2+c^2}{9}$, $\text{Pow}(H) = -8R^2 \cos A \cos B \cos C$ (acute); $AH \cdot HD = 4R^2 \cos A \cos B \cos C$.

Quadrilaterals

Parallelogram law: $2(a^2 + b^2) = p^2 + q^2$. **Euler**: $a^2 + b^2 + c^2 + d^2 = p^2 + q^2 + 4m^2$ (m = distance between diagonal midpoints). **British flag**: P and rectangle $ABCD$: $PA^2 + PC^2 = PB^2 + PD^2$. Perpendicular diagonals $\iff a^2 + c^2 = b^2 + d^2$. Tangential quadrilateral $\iff a + c = b + d$ (Pitot).

Solids

Euler $V - E + F = 2$. Platonic (V, E, F) : tetrahedron $(4, 6, 4)$, cube $(8, 12, 6)$, octahedron $(6, 12, 8)$, dodecahedron $(20, 30, 12)$, icosahedron $(12, 30, 20)$. Regular tetrahedron volume $\frac{a^3}{6\sqrt{2}}$. Spherical cap area $2\pi Rh$. n -ball volume $\frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)} R^n$.

MORE COUNTING

Formulas

$S(n, k) = \frac{1}{k!} \sum_j (-1)^j \binom{k}{j} (k-j)^n$. Narayana $N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n-1}{k-1}$, $\sum_k N(n, k) = C_n$.

Positive solutions of $x_1 + \cdots + x_k = n$: $\binom{n-1}{k-1}$; compositions of n : 2^{n-1} ; multisets of size k from n types: $\binom{n+k-1}{k}$.

k -subsets of $[n]$ with no two consecutive: $\binom{n-k+1}{k}$.

Partitions into exactly k parts: $p(n, k) = p(n-1, k-1) + p(n-k, k)$; into at most k parts = into parts $\leq k$ (conjugate the Young diagram).

Involutions $I_n = I_{n-1} + (n-1)I_{n-2}$; perfect matchings of K_{2n} : $(2n-1)!!$; necklaces with n beads, k colours: $\frac{1}{n} \sum_{d|n} \varphi(d) k^{n/d}$.

Chromatic polynomials: tree $k(k-1)^{n-1}$; cycle $(k-1)^n + (-1)^n (k-1)$; K_n : $k(k-1) \cdots (k-n+1)$. Hypercube Q_n : 2^n vertices, $n2^{n-1}$ edges.

More probability

Max of n uniforms has mean $\frac{n}{n+1}$; expected records in a random permutation H_n ; birthday: $P(\text{no match}) \approx e^{-n^2/(2N)}$. **Chernoff**: sum of independent Bernoullis with mean μ : $P(X \geq (1+\delta)\mu) \leq e^{-\delta^2 \mu/3}$ ($0 < \delta \leq 1$).

Total expectation $E[X] = E[E[X | Y]]$; Wald $E[\sum_{i \leq N} X_i] = E[N]E[X]$ (N a stopping time); $E[XY]^2 \leq E[X^2]E[Y^2]$. Simple random walk on $\mathbb{Z}$ and $\mathbb{Z}^2$ is recurrent, on $\mathbb{Z}^3$ transient.

MORE INTEGRALS, SERIES, PRODUCTS

Definite integrals

$$\begin{aligned} \int_{-\infty}^{\infty} x^{2n} e^{-x^2} dx &= \frac{(2n-1)!!}{2^n} \sqrt{\pi}; \int_0^{\infty} x^n e^{-x} dx = n!; \int_0^{\infty} e^{-ax} \sin bx dx = \frac{b}{a^2+b^2}; \\ \int_0^{\infty} \frac{\sin^2 x}{x^2} dx &= \frac{\pi}{2}; \int_0^{\infty} \frac{dx}{1+x^n} = \frac{\pi/n}{\sin(\pi/n)}; \int_0^{\infty} e^{-x^2} \cos 2bx dx = \frac{\sqrt{\pi}}{2} e^{-b^2}; \\ \int_0^1 \frac{\ln(1-x)}{x} dx &= -\frac{\pi^2}{6}; \int_0^1 \ln x \ln(1-x) dx = 2 - \frac{\pi^2}{6}; \int_0^1 x^{-x} dx = \sum_{n \geq 1} n^{-n}; \\ \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx &= \frac{\pi^2}{4} \text{ (use } x \mapsto \pi-x); \int_0^{\infty} \frac{\ln(1+x^2)}{1+x^2} dx = \pi \ln 2; \\ \int_0^{\pi/2} \sin^m x \cos^n x dx &= \frac{1}{2} B\left(\frac{m+1}{2}, \frac{n+1}{2}\right). \end{aligned}$$

Series

$$\begin{aligned} \sum_{n \geq 1} \frac{(-1)^{n+1}}{n^2} &= \frac{\pi^2}{12}; \sum \frac{1}{4n^2-1} = \frac{1}{2}; \sum \frac{n^2}{2^n} = 6; \sum \frac{n}{(n+1)!} = 1; \sum \frac{F_n}{2^n} = 2; \\ \sum \frac{H_n}{2^n} &= 2 \ln 2 \text{ (from } \sum H_n x^n = \frac{-\ln(1-x)}{1-x}); \sum \frac{H_n}{n^2} = 2\zeta(3); \\ \sum_{n \geq 1} \frac{1}{n^2 \binom{2n}{n}} &= \frac{\pi^2}{18}; \sum_{n \geq 1} \frac{1}{n^2+1} = \frac{\pi \coth \pi - 1}{2}. \\ \text{Radius of convergence } \frac{1}{\limsup |a_n|^{1/n}}. &\text{ Raabe: } n\left(\frac{a_n}{a_{n+1}} - 1\right) \rightarrow L > 1 \Rightarrow \text{converges.} \end{aligned}$$

Infinite products

$$\begin{aligned} \prod_{n \geq 2} \left(1 - \frac{1}{n^2}\right) &= \frac{1}{2}; \prod_{n \geq 2} \frac{n^3-1}{n^3+1} = \frac{2}{3}; \sin x = x \prod_{n \geq 1} \left(1 - \frac{x^2}{n^2 \pi^2}\right); \\ \text{Viète } \frac{2}{\pi} &= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \dots \prod (1+a_n) \text{ converges} \\ (a_n \geq 0) &\text{ iff } \sum a_n \text{ does.} \end{aligned}$$

MORE LINEAR & ABSTRACT ALGEBRA

Spectral facts

$p_A(\lambda) = \lambda^n - (\text{tr } A)\lambda^{n-1} + \dots + (-1)^n \det A$; $\text{tr } A^k = \sum \lambda_i^k$.
Railey: $\lambda_{\min} \leq \frac{x^T A x}{x^T x} \leq \lambda_{\max}$ (A symmetric). **Gershgorin:** each eigenvalue lies in some disc $|z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}|$. **Perron–Frobenius:** a positive matrix has a simple largest eigenvalue with a positive eigenvector. **Cauchy interlacing** for principal submatrices of symmetric matrices.
Sylvester criterion: $A \succ 0 \iff$ all leading minors > 0 . Hadamard (PSD): $\det A \leq \prod a_{ii}$.
 $\text{rank}(A^T A) = \text{rank } A$ (real); skew-symmetric of odd size $\Rightarrow \det = 0$;
 $A \text{ adj}(A) = (\det A)I$; $|\text{tr}(AB)| \leq \|A\|_F \|B\|_F$.

Group & ring facts

$|S_n| = n!$, $|A_n| = \frac{n!}{2}$, $|D_n| = 2n$, $|(\mathbb{Z}/n)^\times| = \varphi(n)$. Orders p , p^2 : abelian; order $2p$ (p odd): cyclic or dihedral.
Sylow: $|G| = p^a m$, $p \nmid m$: Sylow p -subgroups exist, are conjugate, $n_p \equiv 1 \pmod{p}$, $n_p \mid m$.
Wilson for groups: the product of all elements of a finite abelian group is e , unless there is exactly one element of order 2, in which case it is that element.
 Primitive roots mod p : $\varphi(p-1)$ of them; quadratic residues: $\frac{p-1}{2}$.
 $\mathbb{Z}[i]$: norm $N(a+bi) = a^2 + b^2$ is multiplicative; $\mathbb{Z}[i]$ is a UFD.

MORE SMALL-NUMBER TABLES

Smallest primitive root g mod p :

p	3	5	7	11	13	17	19	23	29	31	37
g	2	2	3	2	2	3	2	5	2	3	2
p	41	43	47	53	59	61	67	71	73	79	83
g	6	3	5	2	2	2	2	7	5	3	2

Harmonic numbers: $H_1..H_{10} = 1, \frac{3}{2}, \frac{11}{6}, \frac{25}{12}, \frac{137}{60}, \frac{49}{20}, \frac{363}{140}, \frac{761}{280}, \frac{7129}{2520}, \frac{7381}{2520}$.

Primitive Pythagorean triples: $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(7, 24, 25)$, $(20, 21, 29)$, $(12, 35, 37)$, $(9, 40, 41)$, $(28, 45, 53)$, $(11, 60, 61)$, $(33, 56, 65)$.

Perfect numbers: 6, 28, 496, 8128. **Mersenne prime exponents:** 2, 3, 5, 7, 13, 17, 19, 31.

Powers: $3^5 = 243$, $3^{10} = 59049$, $5^5 = 3125$, $7^4 = 2401$, $2^{16} = 65536$, $11^3 = 1331$, $12^3 = 1728$, $1729 = 1^3 + 12^3 = 9^3 + 10^3$.

X PRACTICE & PLAN

problems to try, grouped by the idea they train

PROBLEM LADDER BY TECHNIQUE

Problems are named by competition, year and number; statements are easy to find (AoPS, official shortlists). Try each for an hour before reading solutions. **Bold** = worked in this booklet.

■ Inequalities

- **IMO 1995/2** ($abc = 1$; reciprocal substitution, Titu).
- **IMO 2000/2** ($abc = 1$; substitution $a = \frac{x}{y}$).
- **IMO 2001/2** (isolated fudging or Hölder).
- **IMO 2005/3** ($xyz \geq 1$; clever comparison).
- **IMO 2008/2** (squares of $\frac{x}{x-1}$; substitution, then a square).
- **Poland 1996** (tangent line).

■ Functional equations & polynomials

- **IMO 2019/1** (linearisation over $\mathbb{Z}$).
- **IMO 2012/4** ($a + b + c = 0$; values at small integers, parity).
- **IMO 2009/5** (triangle condition; injectivity, periodicity).
- **IMO 2010/3** ($(g(m) + n)(g(n) + m)$ square; primes and differences).
- **IMO 2006/5** (iterated integer polynomials).
- **USAMO 1975** (interpolation).

■ Number theory

- **IMO 1988/6**, **IMO 2007/5** (Vieta jumping).
- **IMO 1990/3** (orders, LTE).
- **IMO 2013/1** (constructive induction on the exponent).
- **IMO 2015/2** (powers of two; size + parity case analysis).
- **IMO 2016/4** (“fragrant” sets for $n^2 + n + 1$; gcds, CRT).

■ Combinatorics

- **IMO 1998/2** (double counting).
- **IMO 1986/3** (quadratic monovariant).
- **IMO 2011/2** (windmill; rotating line invariant).
- **IMO 2014/1** (discrete continuity / monotone sequence).
- **IMO 2017/1** (sequence mod 3, squares; case analysis on residues).

■ Geometry

- **IMO 2006/1** (incenter–excenter lemma).
- **IMO 2008/1** (radical axis).

■ Putnam

- **1980 A3**, **2005 A5** (integral symmetry).
- **1992 A6** (geometric probability; antipodal pairing).
- Any $A1/B1$ from the last 15 years under a 30-minute limit.

A 12-WEEK PLAN

weeks	focus
1–2	Algebra: AM-GM, C-S, substitutions; 5 inequalities + 3 FEs per week
3–4	Number theory: orders, LTE, Vieta jumping, Diophantine methods
5–6	Combinatorics: pigeonhole, invariants, double counting, graphs
7–8	Geometry: angle chasing, power of a point, the lemma list; one bash per week
9	Putnam analysis: limits, integrals, series (if targeting Putnam)
10	Linear algebra, abstract algebra, probability (Putnam) / advanced geometry (IMO)
11–12	Mixed timed sets; two full mock papers per week with write-ups

Each week: one post-mortem page summarising every idea you met, filed into your lemma book under the section headings of this booklet. By week 12 you will have your own version of this cheat sheet, and that is the one that matters.

INDEX OF TECHNIQUES

Abel summation II · Cauchy–Davenport II · Cauchy–Schwarz II, VI · Ceva/Menelaus V · Colouring IV · Combinatorial Nullstellensatz II · Complex bash V · CRT III · Descent III · Determinants mod 2 VI · Discrete continuity I · Double counting IV · Extremal principle I, IV · Generating functions IV · Hall IV · Harmonic bundles V · Hölder II · Homothety V · Incenter–excenter V · Invariants I, IV · Inversion V · Isolated fudging II · Jensen II · King’s rule VI · Kummer/Legendre III · LTE III · Linearity of expectation IV, VI · Miquel V · Orders III · Pigeonhole IV · Power of a point V · Probabilistic method IV · Quadratic residues III · Radical axis V · Roots of unity II, IV · Schur II · SOS II · Stolz–Cesàro VI · Strategy stealing IV · Tangent line II · Thue’s lemma III · uvw II · Vieta jumping III · Burnside IV, VII · Pell III, VII · Simson line V, VII · Monovariant I, VII · Zsigmondy III.